Chapter 4
Local Area network

第四章 局域网

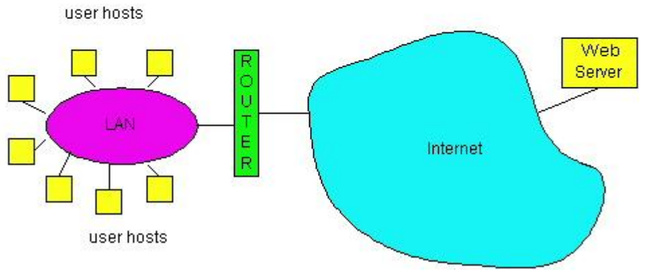
LAN technologies

Data link layer so far:

- services, error detection/correction, multiple access

Next: LAN technologies

- LAN model
- addressing
- Ethernet
- hubs, switches
- 802.11
- 802.15



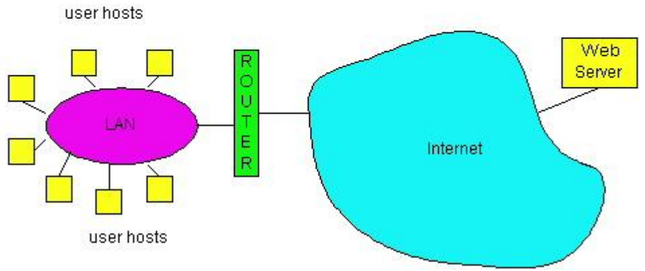
局域网技术

数据链路层至今：

- 服务，错误检测/纠正，多路访问

下一节：局域网技术

- 局域网模型
- 地址
- 以太网
- 集线器、交换机
- 802.11
- 802.15



Keypoints and Difficulties

Keypoints:

- LAN model
- Ethernet
- Hubs, switches
- Wireless LAN- IEEE 802.11

Difficulties:

- The minimum frame length
- The exponential Backoff algorithm
- CSMA/CA

关键点和难点

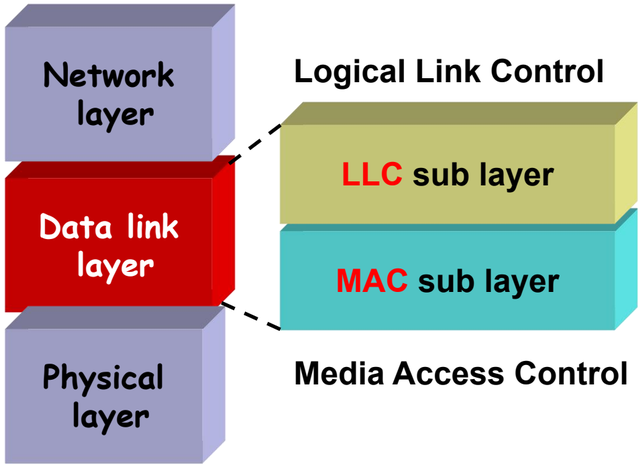
关键点：

- 局域网模型
- 以太网
- 集线器、交换机
- 无线局域网- IEEE 802.11

难点：

- 最小帧长度
- 指数退避算法
- CSMA/CA

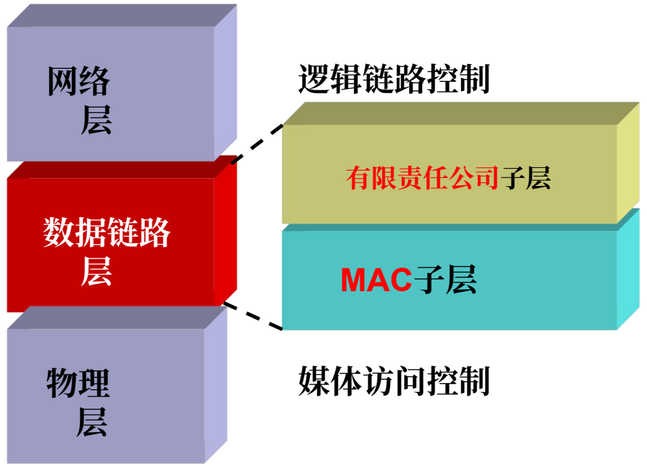
LAN model



Provide the connection environment to the upper level

Providing media access to the lower layer

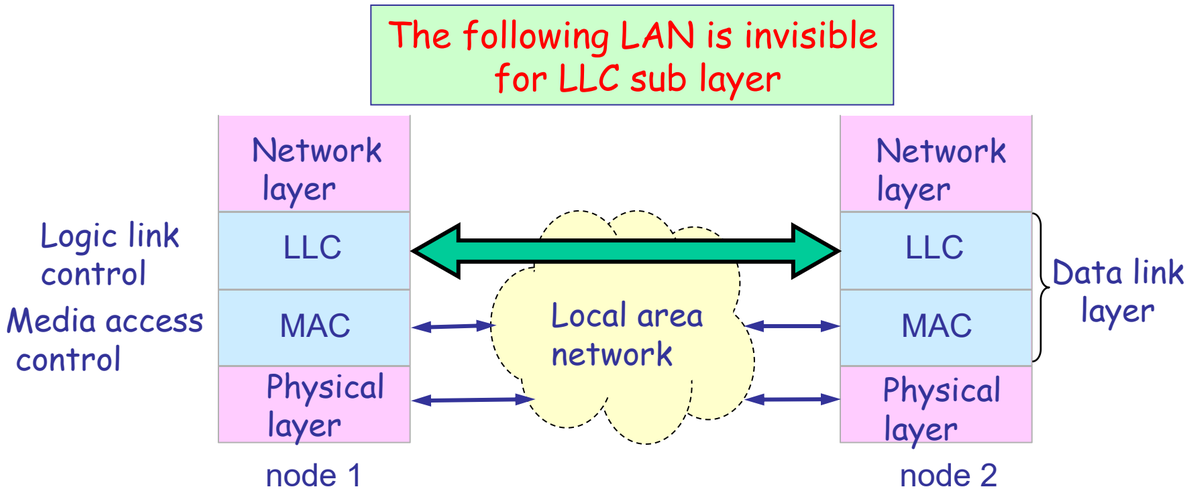
局域网模型



提供连接环境到上层级别

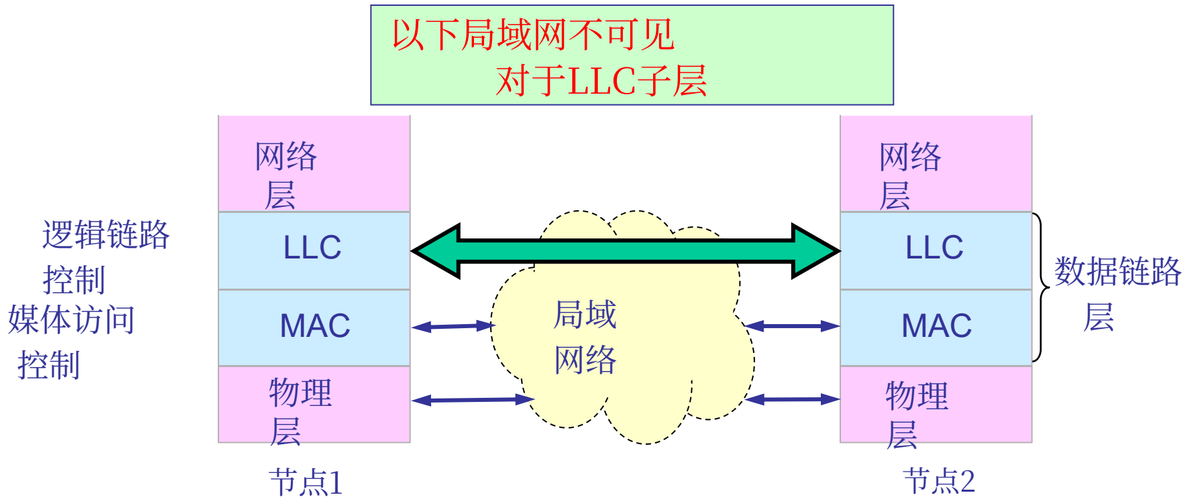
提供媒体访问给下层

LAN model



For the same LLC, several MAC options may be provided.

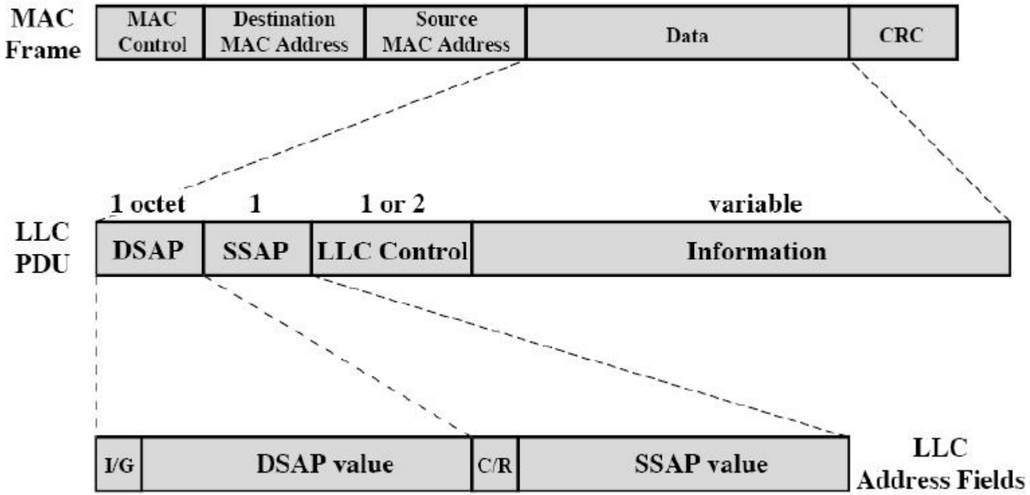
局域网模型



对于同一个有限责任公司，可能会提供多个MAC选项。

LLC and MAC

MAC Frame Format

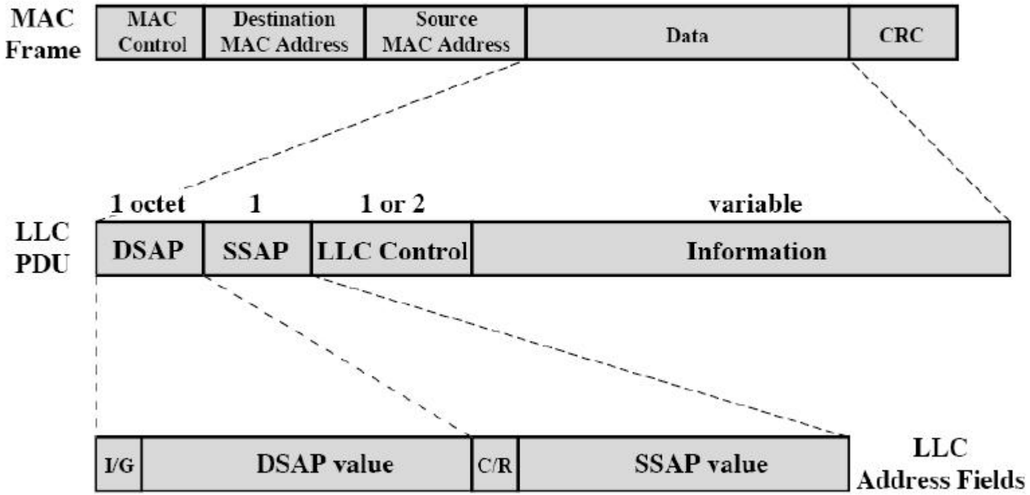


I/G = individual/group
C/R = command/response

DSAP = destination service access point
SSAP = source service access point

有限责任公司和

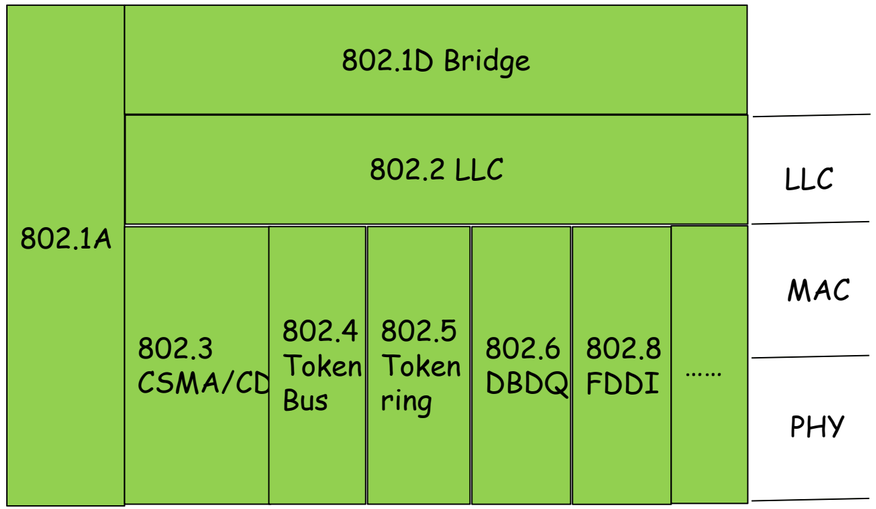
MAC Frame Format



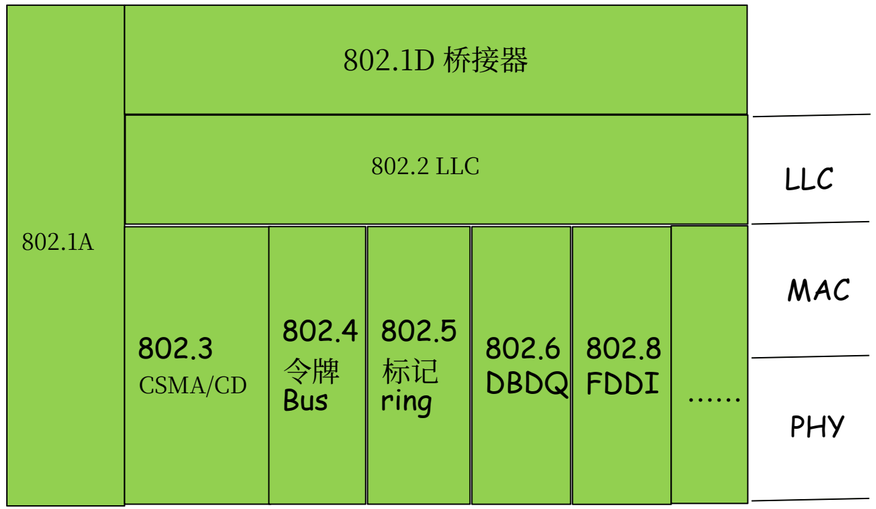
I/G = individual/group
C/R = command/response

DSAP = destination service access point
SSAP = source service access point

IEEE 802 working group



IEEE 802 工作组



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LAN Addresses

32-bit IP address:

- *network-layer* address
- used to get datagram to destination network

LAN (or MAC or physical) address:

- used to get data frame from one interface to another physically-connected interface (same network)
- 48 bit MAC address (for most LANs) burned in the adapter ROM

局域网地址

32位IP地址：

- 网络层地址
- 用于将数据报传送到目标网络

局域网（或MAC或物理）地址：

- 用于将数据帧从一个接口传输到另一个物理连接的接口（同一网络）

- 48位MAC地址（大多数局域网使用）烧录在适配器ROM中

LAN Address (more)

- MAC address allocation administered by IEEE
- manufacturer buys portion of MAC address space (to assure uniqueness)
- Analogy:
  - (a) MAC address: like Social Security Number
  - (b) IP address: like postal address
- MAC flat address => portability
  - can move LAN card from one LAN to another
- IP hierarchical address NOT portable
  - depends on network to which one attaches

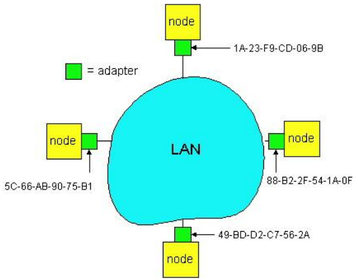
局域网地址（更多）

- MAC地址分配由IEEE管理
- 制造商购买MAC地址空间的一部分（以确保唯一性）

- 类比：
  - (a) MAC地址：类似于社会安全号码
  - (b) IP地址：类似于邮政地址
- MAC平面地址 => 可移植性
  - 可以将局域网卡从一个局域网移动到另一个局域网
- IP层级地址不可移植
  - 取决于您连接的网络

LAN Address

Each adapter on LAN has unique LAN address





IEEE 802局域网的MAC地址格式

扩展的唯一标识符EUI
EUI-48

组织唯一标识符OUI（由IEEE的注册管理机构分配）								网络接口标识符（由获得OUI的厂商自行随意分配）							
第一字节				第二字节				第三字节				第四字节			
b7 b6 b5 b4 b3 b2 b1 b0				b7 b6 b5 b4 b3 b2 b1 b0				b7 b6 b5 b4 b3 b2 b1 b0				b7 b6 b5 b4 b3 b2 b1 b0			
十六进制	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

标准表示法：XX-XX-XX-XX-XX-XX

例如：00-0C-CF-93-8C-92

其他表示法：XX:XX:XX:XX:XX:XX

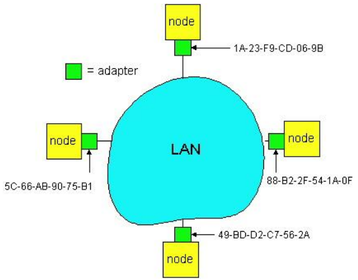
例如：00:0C:CF:93:8C:92

XXXX.XXXX.XXXX

例如：000C.CF93.8C92

局域网地址

每个适配器在局域网上有唯一的局域网地址





IEEE 802局域网的MAC地址格式

扩展的唯一标识符EUI
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十六进制	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

标准表示法：XX-XX-XX-XX-XX-XX

例如：00-0C-CF-93-8C-92

其他表示法：XX:XX:XX:XX:XX:XX

例如：00:0C:CF:93:8C:92

XXXX.XXXX.XXXX

例如：000C.CF93.8C92

MAC Addresses

IEEE 802局域网的MAC地址格式

扩展的唯一标识符EUI
EUI-48



第一字节的b1位	第一字节的b0位	MAC地址类型	地址数量占比	总地址数量
0	0	全球管理 单播地址 厂商生产网络设备（网卡，交换机，路由器）时固化	1/4	2^{48}=281,474,976,710,656（二百八十多万亿）
	1	全球管理 多播地址 标准网络设备所支持的多播地址，用于特定功能	1/4	
1	0	本地管理 单播地址 由网络管理员分配，覆盖网络接口的全球管理单播地址	1/4	
	1	本地管理 多播地址 用户对主机进行软件配置，以表明其属于哪些多播组 注意：剩余46位全为1时，就是广播地址FF-FF-FF-FF-FF-FF	1/4	

https://standards-oui.ieee.org/oui/oui.txt
MAC地址查询 - https://mac.bmcx.com/

What is the random MAC address technology?

MAC地址

IEEE 802局域网的MAC地址格式

扩展的唯一标识符EUI
EUI-48



第一字节的b1位	第一字节的b0位	MAC地址类型	地址数量占比	总地址数量
0	0	全球管理 单播地址 厂商生产网络设备（网卡，交换机，路由器）时固化	1/4	2^{48}=281,474,976,710,656（二百八十多万亿）
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	1	本地管理 多播地址 用户对主机进行软件配置，以表明其属于哪些多播组 注意：剩余46位全为1时，就是广播地址FF-FF-FF-FF-FF-FF	1/4	

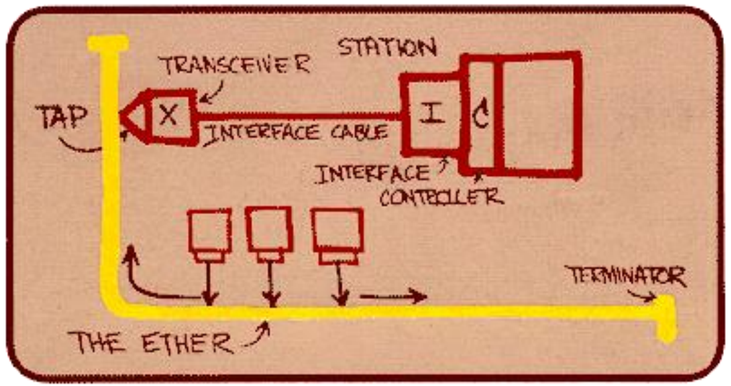
https://standards-oui.ieee.org/oui/oui.txt
MAC地址查询 - https://mac.bmcx.com/

什么是随机的MAC地址技术?

Ethernet

"dominant" LAN technology:

- cheap $20 for 100Mbs!
- first wildey used LAN technology
- Simpler, cheaper than token LANs and ATM
- Kept up with speed race: 10, 100, 1000 Mbps

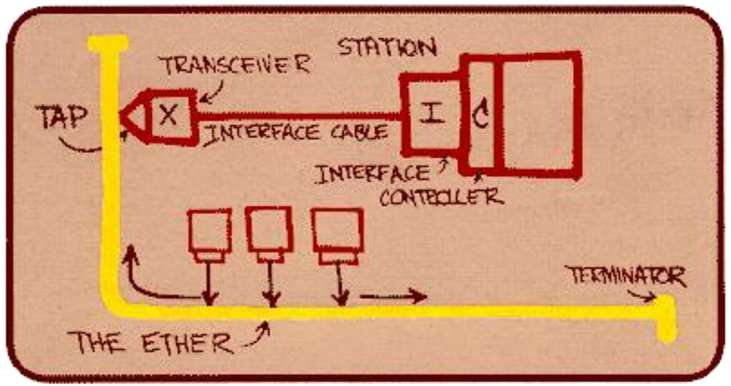


Metcalfe's Etheret sketch

以太网

"主导"局域网技术:

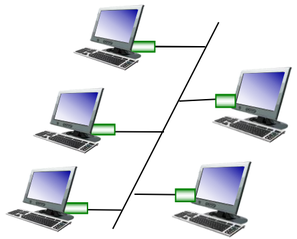
- 便宜 $20 可达 100Mbps!
- 最早使用的局域网技术
- 比令牌局域网和 ATM 更简单、更便宜
- 紧跟速度竞赛:10、100、1000 Mbps



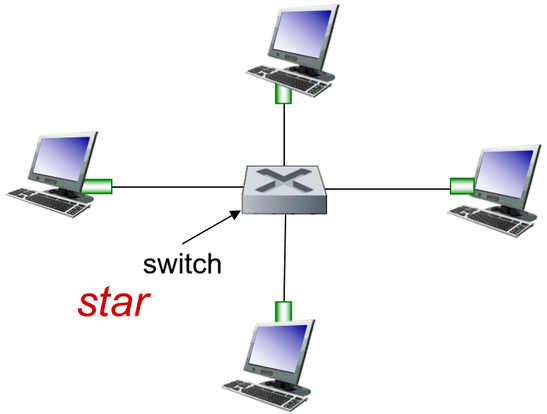
Metcalfe的以太网草图

Ethernet: physical topology

- *bus:* popular through mid 90s
  - all nodes in same collision domain (can collide with each other)
- *star:* prevails today
  - active *switch* in center
  - each "spoke" runs a (separate) Ethernet protocol (nodes do not collide with each other)



bus: coaxial cable



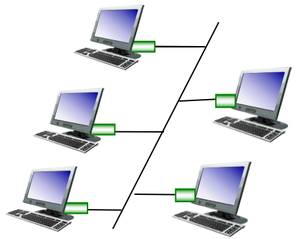
star

以太网：物理拓扑

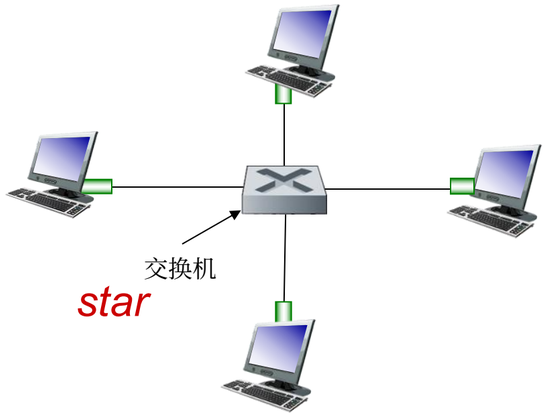
- 总线： 在90年代中期非常流行 ○小节点在同一冲突域（可以相互冲突）
- 星型： 目前占主导地位 ○主动

交换机 位于中心

每 个“辐条”运行一个（独立的）以太网协议（节点之间不会发生冲突）



总线： 同轴电缆



star

Ethernet: unreliable, connectionless

- *connectionless:* no handshaking between sending and receiving NICs
- *unreliable:* receiving NIC doesn't send acks or nacks to sending NIC
  - data in dropped frames recovered only if initial sender uses higher layer rdt (e.g., TCP), otherwise dropped data lost
- Ethernet's MAC protocol: unslotted *CSMA/CD with binary backoff*

以太网：不可靠，无连接

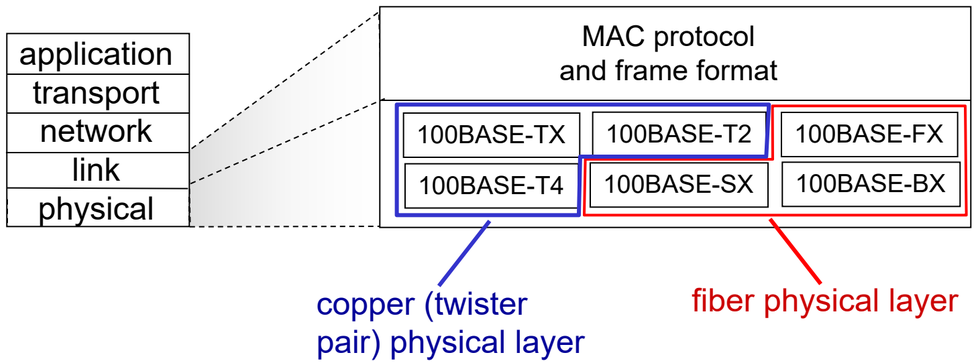
无 无连接： 发送和接收网卡之间没有握手

- 不可靠： 接收网卡不向发送网卡发送 ACK 或 NAK
  - 丢弃帧中的数据只有在初始发送者使用高层 rdt（例如 TCP）时才能恢复，否则丢失的数据将永久丢失
- 以太网's MAC 协议：无时隙 *CS MA/CD* 带二进制退避

4-14

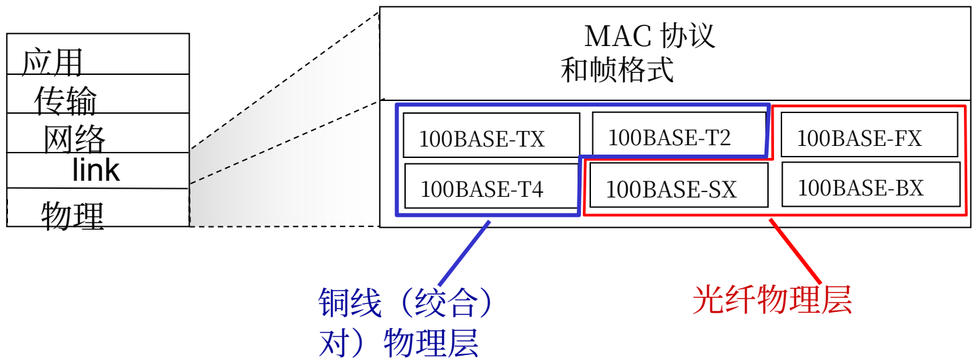
802.3 Ethernet standards: link & physical layers

- *many* different Ethernet standards
  - common MAC protocol and frame format
  - different speeds: 10 Mbps, 100 Mbps, 1Gbps, 10 Gbps, 40 Gbps
  - different physical layer media: fiber, cable

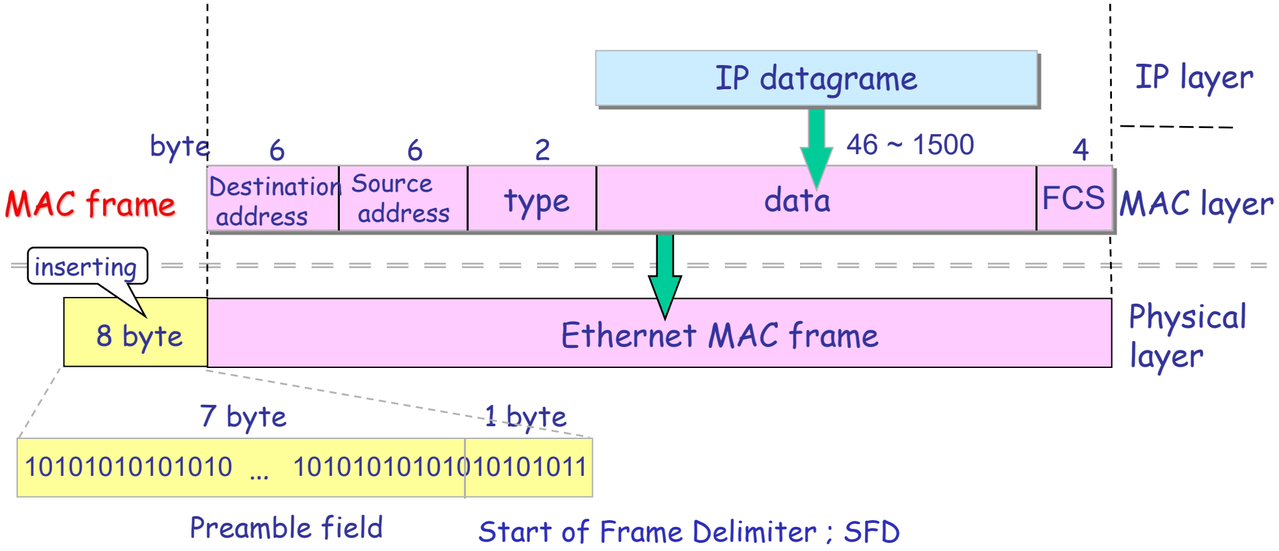


802.3 以太网标准：链路和物理层

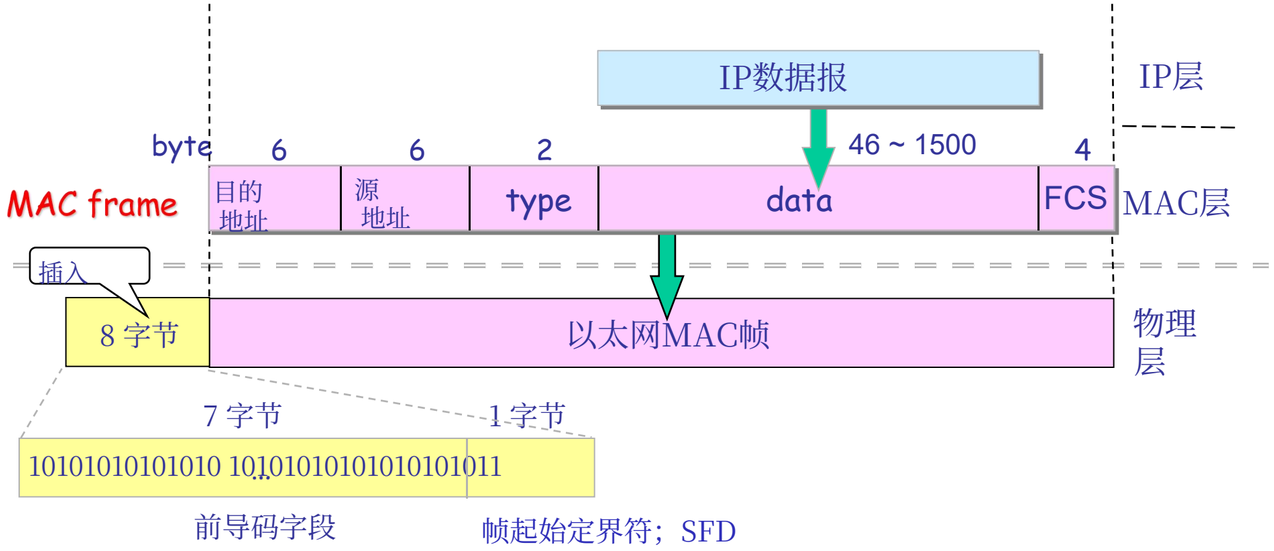
- 许多 不同的以太网标准
  - 常见的 MAC 协议和帧格式
  - 不同的速率：10 Mbps、100 Mbps、1 Gbps、10 Gbps、40 Gbps
  - 不同的物理层介质：光纤、电缆



Ethernet Frame Structure



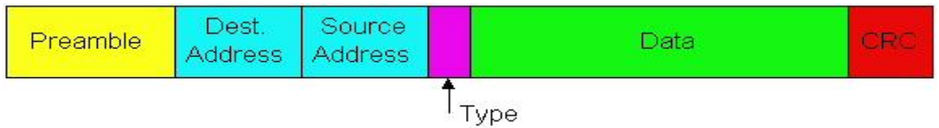
以太网帧结构



4-16

Ethernet Frame Structure

Sending adapter encapsulates IP datagram (or other network layer protocol packet) in Ethernet frame

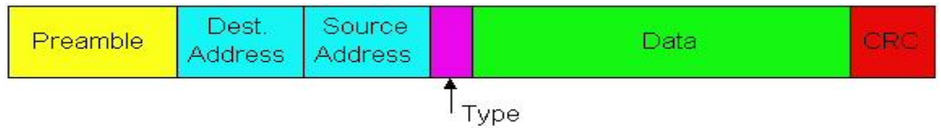


Preamble:

- 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- used to synchronize receiver, sender clock rates

以太网帧结构

发送适配器将IP数据报（或其他网络层协议数据包）封装在以太网帧

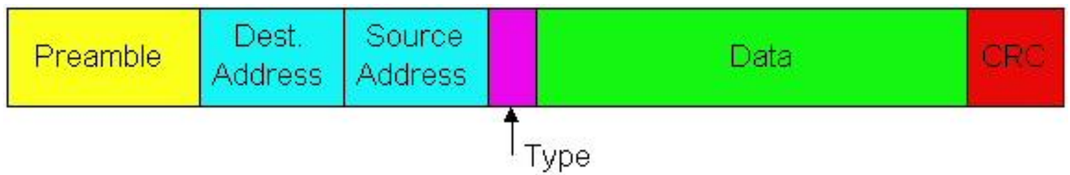


前导码：

- 7个字节，模式为10101010，随后跟一个模式为10101011的字节
- 用于同步接收器和发送器的时钟速率

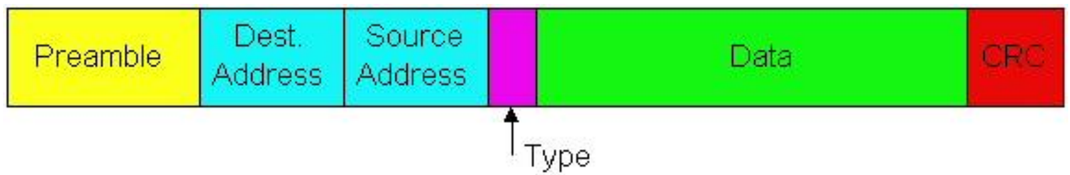
Ethernet Frame Structure (more)

- Addresses: 6 bytes, frame is received by all adapters on a LAN and dropped if address does not match
- Type: 2 bytes, indicates the higher layer protocol, mostly IP but others may be supported such as Novell IPX and AppleTalk)
- CRC: 4 bytes, checked at receiver, if error is detected, the frame is simply dropped



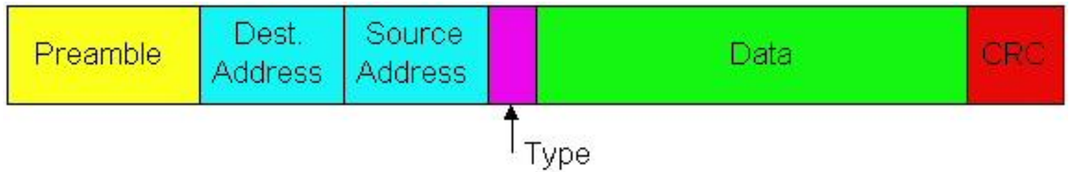
以太网帧结构 (更多)

- 地址： 6 字节，帧会被局域网上的所有适配器接收，如果地址不匹配则会被丢弃
- 类型： 2 字节，表示高层协议，主要是 IP，但也可能支持其他协议，如 Novell IPX 和 AppleTalk)
- CRC： 4 字节，在接收端进行校验，如果检测到错误，帧将被简单地丢弃



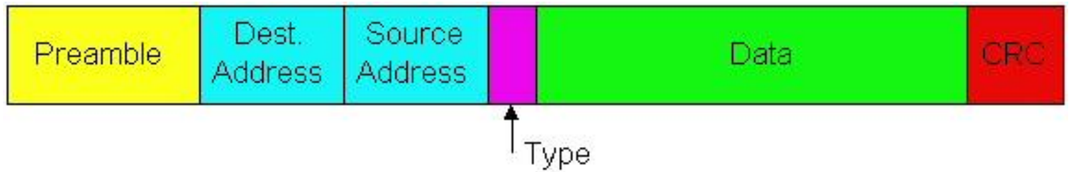
Ethernet Frame Structure (more)

- Data: 46~1500 bytes
- Minimum frame length: 64 bytes, why? (contention period 2τ is 51.2 μs for IEEE 802.3, R=10Mbps)
- Maximum frame length: 1518 bytes, why?



以太网帧结构(更多)

- 数据： 46~1500 字节
- 最小帧长度： 64 字节，为什么？(竞争周期 2τ 是 51.2 μs，对于 IEEE 802.3，R=10Mbps)
- 最大帧长度： 1518 字节，为什么？



Exercises-1

- In a LAN using CSMA / CD protocol, the transmission medium is a complete cable, the transmission rate is 1Gbps, and the signal propagation rate in the cable is 200000 km / s. if the minimum data frame length is reduced by 800 bits, the distance between the farthest two stations needs to be at least

(1) increased by 160m (2) increased by 80m
(3) reduced by 160m (4) reduced by 80m

练习-1

- 在使用 CSM A/CD 协议的局域网中，传输介质是一条完整的电缆，传输速率是 1 Gbps，电缆中的信号传播速率是 200000 公里/秒。如果将最小数据帧长度减少 800 比特，那么最远两个站点之间的距离至少需要

(1) 增加 160m (2) 增加 80m (3) 减少 160m (4) 减少 80m

Ethernet: uses CSMA/CD

A: sense channel, if idle

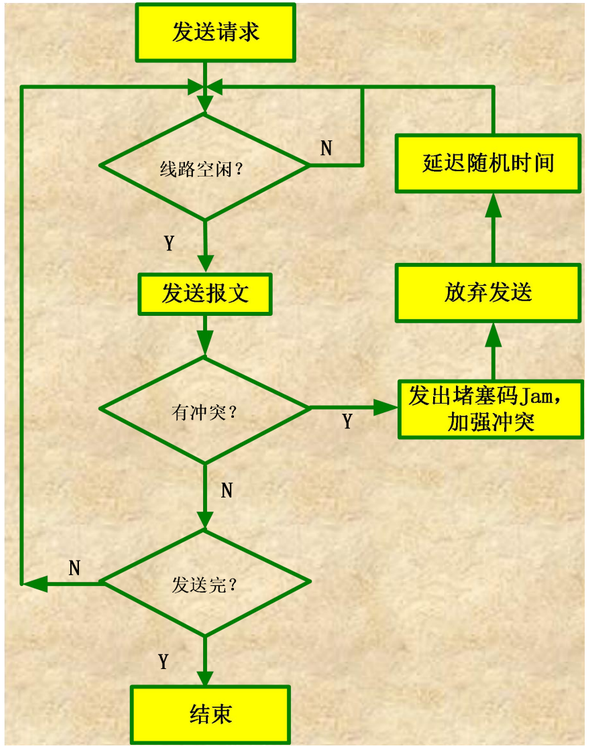
```
then {
        transmit and monitor the channel;
        If detect another transmission
          then {
            abort and send jam signal;
            update # collisions;
            delay as required by exponential backoff algorithm;
            goto A
            }
         else {done with the frame; set collisions to zero}
    }
else {wait until ongoing transmission is over and goto A}
```


以太网：使用CSMA/CD

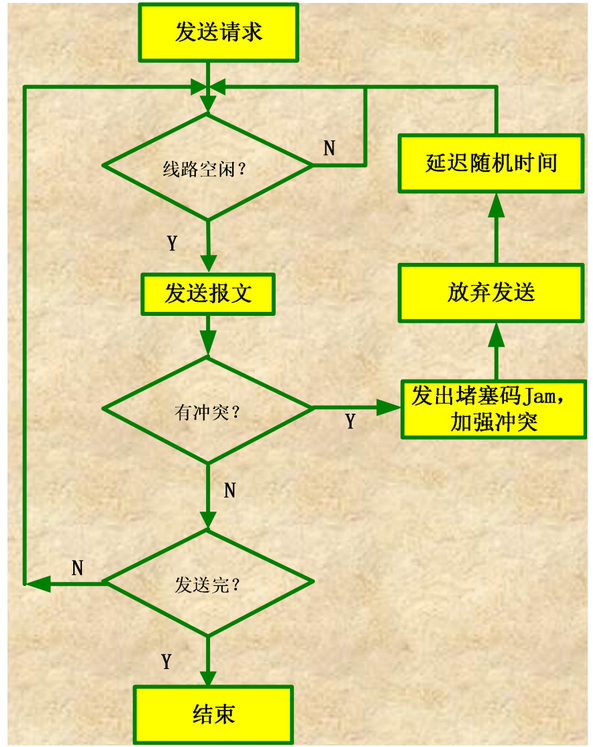
A: 感知信道， 如果空闲则 { 发送并监控信道；如果检测到其他传输则 { 中断并发送阻塞信号；更新冲突次数；按照指数退避算法进行延迟；跳转至 A}否则 { 完成帧传输；将冲突次数设为零} }否则 { 等待当前传输结束并 跳转至 A}

4-21

Ethernet: uses CSMA/CD

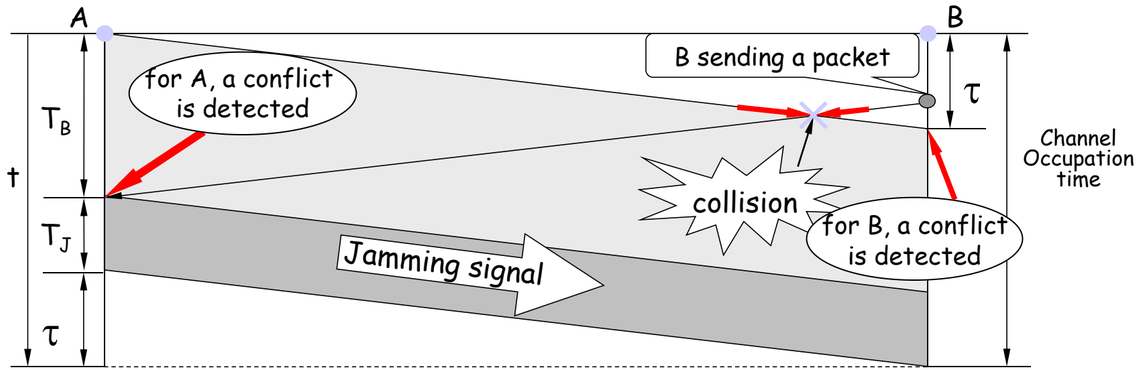


以太网：使用CSMA/CD



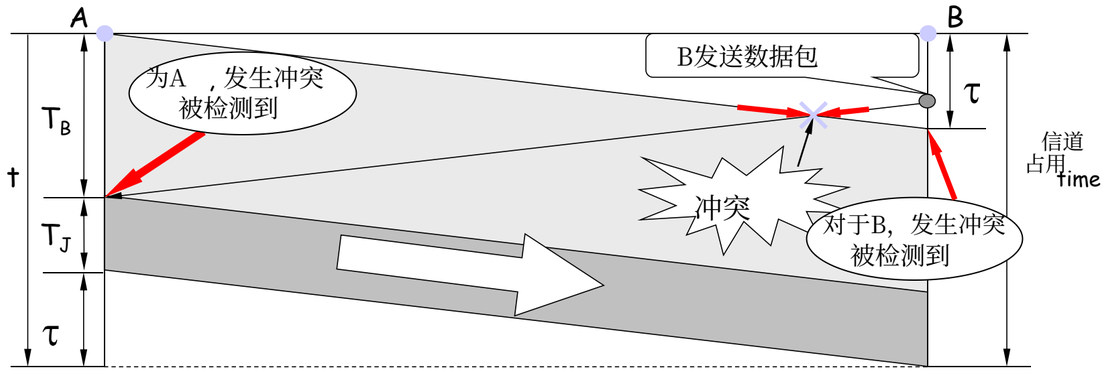
Ethernet's CSMA/CD (more)

Jam Signal: make sure all other transmitters are aware of collision; 48 bits;



以太网的CSMA/CD（更多）

阻塞信号：确保所有其他发射器都意识到冲突；48比特；



Ethernet's CSMA/CD (more)

Exponential Backoff:

- *Goal*: adapt retransmission attemtps to estimated current load
  - heavy load: random wait will be longer
- delay is K x 512 bit transmission times (contention period: 2τ)
- first collision: choose K from {0,1};
- after second collision: choose K from {0,1,2,3}…
- after ten or more collisions, choose K from {0,1,2,3,4,…,1023}
- K=min(n,i),n: # collisions,n≤j(attempt limit);i:back off limit
- For Ethernet, i=10,j=16

以太网的CSMA/CD（更多）

指数退避：

- 目标：根据估计当前负载调整重传尝试
  - 负载严重：随机等待时间会更长
- 延迟是K x 512位传输时间（冲突周期：2τ）
- 第一次冲突：从{0,1}中选择K;
- 第二次冲突后：从{0,1,2,3}中选择K…
- 在十次或更多冲突后，从 {0,1,2,3,4,…,1023} 中选择 K
- K=min(n,i),n: # 冲突,n≤j(尝试限制);i:退避限制
- 对于以太网，i=10,j=16

Interconnecting LANs

Q: Why not just one big LAN?

- Limited amount of supportable traffic: on single LAN, all stations must share bandwidth
- limited length: 802.3 specifies maximum cable length
- large "collision domain" (can collide with many stations)

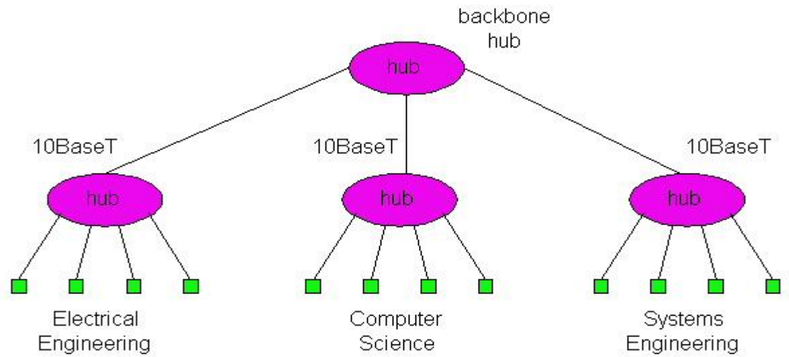
互联局域网

Q: 为什么不只是一个大的局域网?

- 有限的支持流量：在单个局域网中，所有站点必须共享带宽
- 长度有限：802.3规定了最大电缆长度
- 大的“冲突域”(可能与许多站点冲突)

Hubs

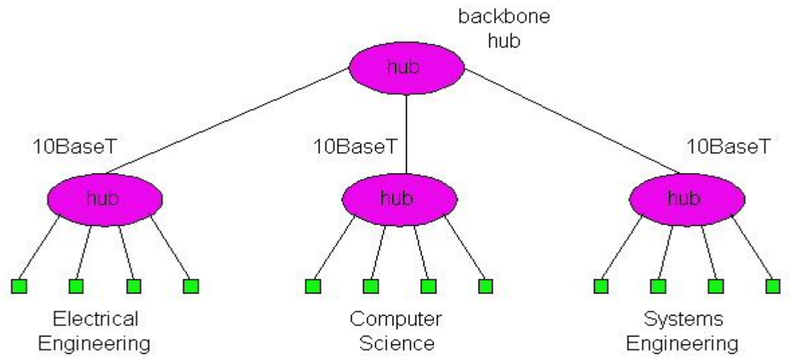
- Physical Layer devices: essentially repeaters operating at bit levels: repeat received bits on one interface to all other interfaces
- Hubs can be arranged in a hierarchy (or multi-tier design), with backbone hub at its top



Hubs

- 物理层设备：本质上是在比特级别运行的重复器：在一个接口上重复接收到的比特，并将其传输到所有其他接口

- 集线器可以按 层次结构（或多层设计）排列，其顶部是骨干集线器



Hubs (more)

- Each connected LAN referred to as LAN **segment**
- Hubs do not isolate collision domains: node may collide with any node residing at any segment in LAN
- Hub Advantages:
  - simple, inexpensive device
  - Multi-tier provides graceful degradation: portions of the LAN continue to operate if one hub malfunctions
  - extends maximum distance between node pairs (100m per Hub)

集线器（更多）

- 每个连接的局域网称为局域网 **段**
- 集线器不会隔离冲突域：节点可能与局域网中任何网段的任何节点发生冲突
- 集线器优势：
  - 简单、价格低廉的设备
  - 多层结构提供优雅降级：如果某个集线器故障，局域网的某些部分仍可继续运行
  - 扩展节点对之间的最大距离（每个Hub为100米）

Hub limitations

- single collision domain results in no increase in max throughput
  - multi-tier throughput same as single segment throughput
- individual LAN restrictions pose limits on number of nodes in same collision domain and on total allowed geographical coverage
- cannot connect different Ethernet types (e.g., 10BaseT and 100baseT)

Hub 限制

- 单个冲突域不会导致最大吞吐量增加
  - 多层吞吐量与单段吞吐量相同
- 单个局域网限制会对同一冲突域中的节点数量和总允许地理覆盖范围造成限制
- 不能连接不同类型的以太网（例如，10BaseT 和 100baseT）

4-28

Ethernet switch

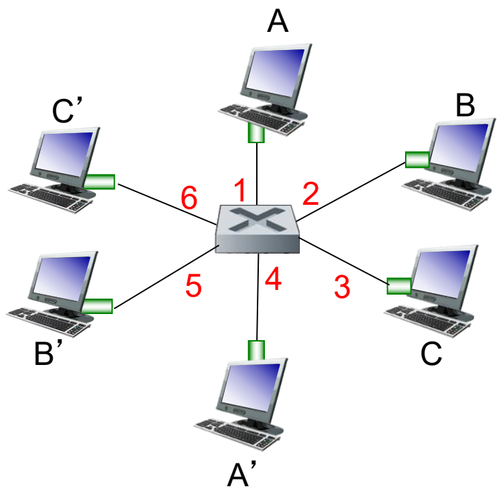
- link-layer device: takes an *active* role
  - store, forward Ethernet frames
  - examine incoming frame's MAC address, selectively forward frame to one-or-more outgoing links when frame is to be forwarded on segment, uses CSMA/CD to access segment
- *transparent*
  - hosts are unaware of presence of switches
- *plug-and-play, self-learning*
  - switches do not need to be configured

以太网交换机

- 数据链路层设备：承担<样式 id='3'>主动</样式>角色
  - 存储转发以太网帧
  - 检查传入的帧'的MAC地址， 有选择地当帧需要在该段上转发时，将帧转发到一个或多个出站链路，使用CSMA/CD访问该段
- 透明
  - 主机不知道交换机的存在 即插即用，自学习
  - 交换机不需要配置

Switch: *multiple* simultaneous transmissions

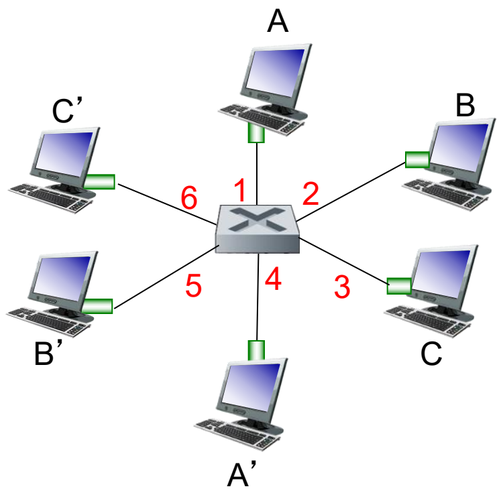
- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, but no collisions; full duplex
  - each link is its own collision domain
- *switching:* A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six interfaces (1,2,3,4,5,6)

交换机：多个同时传输

- 主机有专用的、直接的连接到交换机 h
- 交换机缓存数据包 在每条入站链路上使用以太网协议，但无冲突；全双工
  - 每条链路都是独立的冲突域
- 交换： A到A'和B到B'可以同时传输，无冲突



六接口交换机（*1、2、3、4、5、6*）

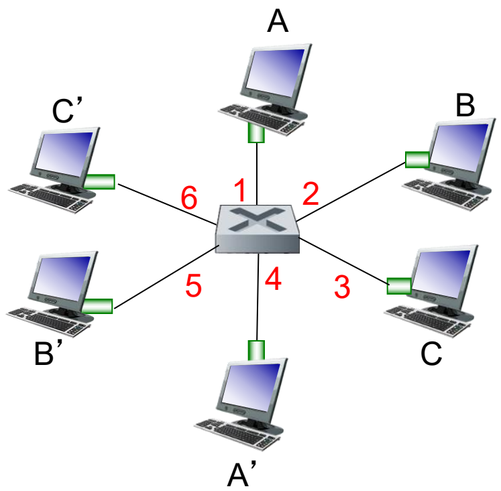
Switch forwarding table

Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

- A: each switch has a switch table, each entry:
  - (MAC address of host, interface to reach host, time stamp)
  - looks like a routing table!

Q: how are entries created, maintained in switch table?

- something like a routing protocol?



switch with six interfaces (1,2,3,4,5,6)

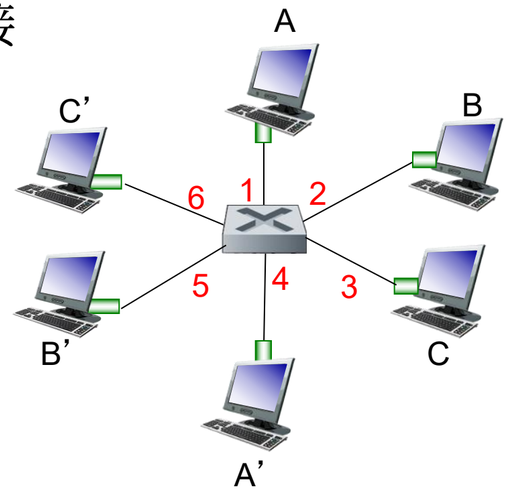
交换机转发表

Q: 交换机如何知道A'通过接口4可达，B'通过接口5可达?

- A: 每个交换机都有一个交换表，每个条目:
  - (主机MAC地址，到达主机的接口，时间戳)
  - 看起来像路由表!

Q: 条目如何在交换表中创建和维护?

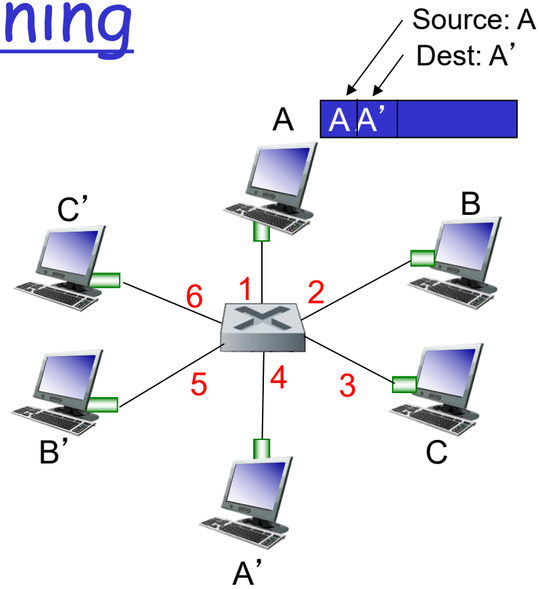
- 类似路由协议的东西?



一个带有六个接口的交换机（1、2、3、4、5、6）

Switch: self-learning

- switch *learns* which hosts can be reached through which interfaces
  - when frame received, switch "learns" location of sender: incoming LAN segment
  - records sender/location pair in switch table

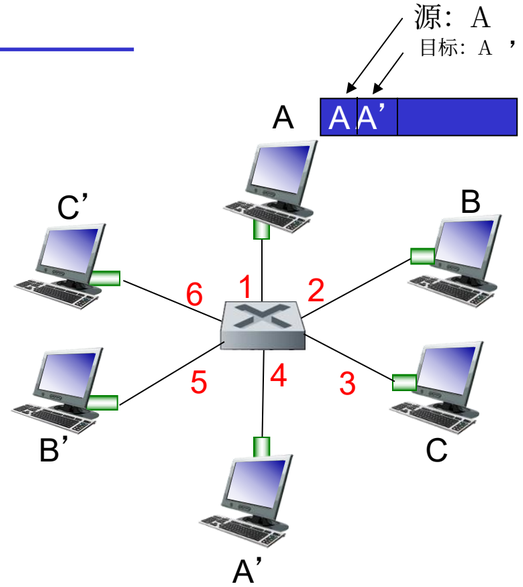


MAC addr	interface	TTL
A	*1*	*60*

Switch table (initially empty)

交换机：自学习

- 交换机 学习 哪些主机可以通过哪些接口被访问
  - 当收到帧时，交换机 "学习" 发送者的位置：传入的局域网段
  - 在交换表中记录发送者/位置对



MAC地址	接口	TTL
A	*1*	*60*

交换表（初始为空）

Switch: frame filtering/forwarding

when frame received at switch:

1. record incoming link, MAC address of sending host
2. index switch table using MAC destination address
3. if entry found for destination
   then {
    if destination on segment from which frame arrived
      then drop frame
      else forward frame on interface indicated by entry
    }
   else flood /* forward on all interfaces except arriving
                 interface */

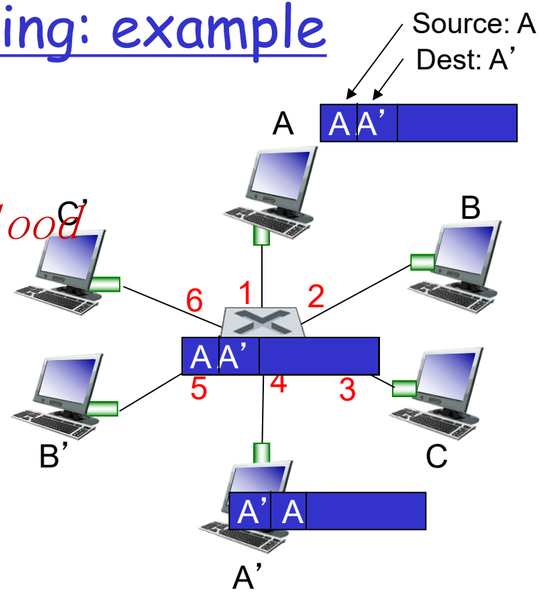
交换机：帧过滤/转发

当帧在交换机接收时：

1. 记录入站链路、发送主机的MAC地址 2. 使用MAC目的地址索引交换表
3. 如果 找到目标地址的条目则 { 如果 目标地址在帧到达的段上则丢弃帧否则通过条目指定的接口转发帧} 否则泛洪 /* 除到达接口外，在所有接口上转发 */

Self-learning, forwarding: example

- frame destination, A', location unknown: *flood*
- destination A location known: selectively send on just one link

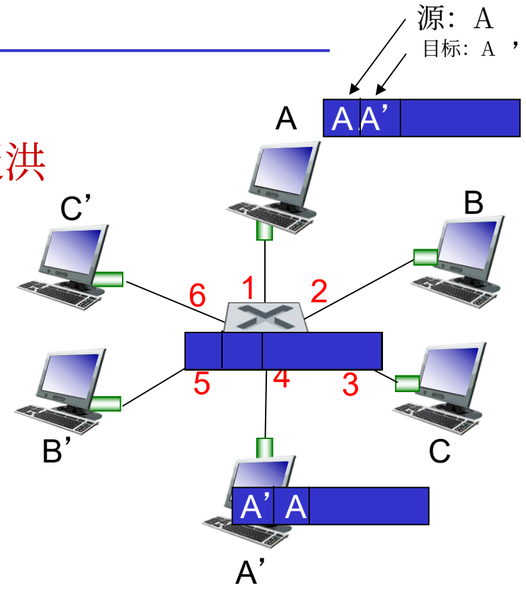


MAC addr	interface	TTL
A	1	60
A'	4	60

switch table (initially empty)

自学习，转发：示例

- 帧目标，A'，位置未知：泛洪
- 目的地 A 已知位置：仅在一个链路上发送

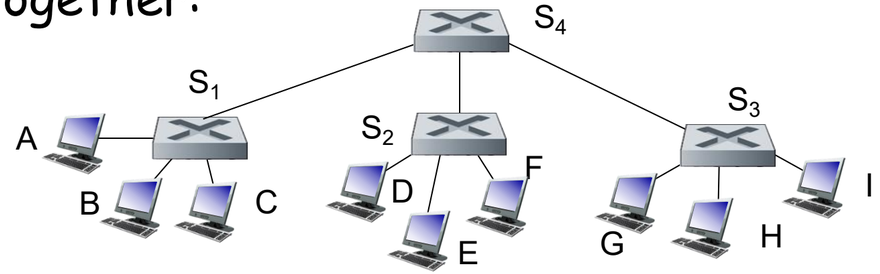


MAC地址	接口	TTL
A	1	60
A'	4	60

交换表（初始为空）

Interconnecting switches

self-learning switches can be connected together:

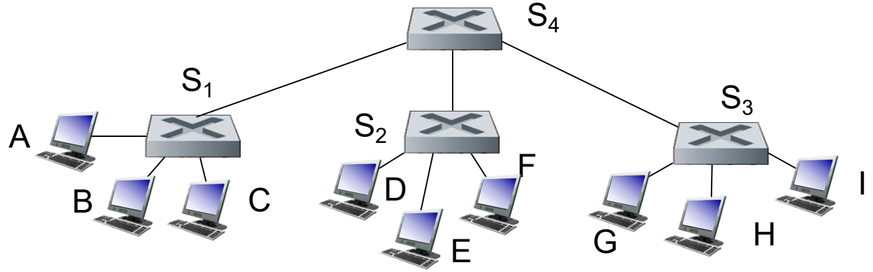


Q: sending from A to G - how does S_1 know to forward frame destined to G via S_4 and S_3?

- A: self learning! (works exactly the same as in single-switch case!)

互联交换机

自学习交换机可以互相连接：

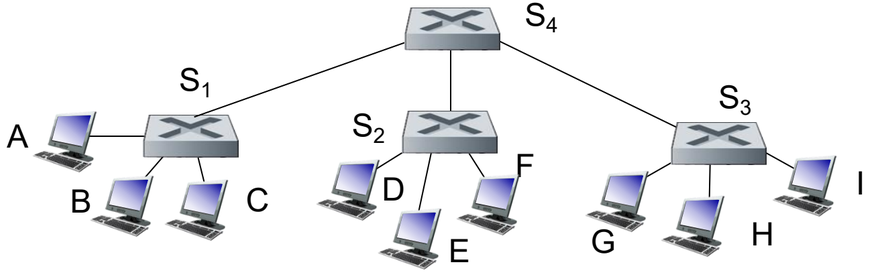


Q：从 A 发送到 G - S 如何知道将发往 G 的帧通过 S_1 转发$_4$ 和 S3?

- A: 自主学习！(与单交换机情况完全相同！)

Self-learning multi-switch example

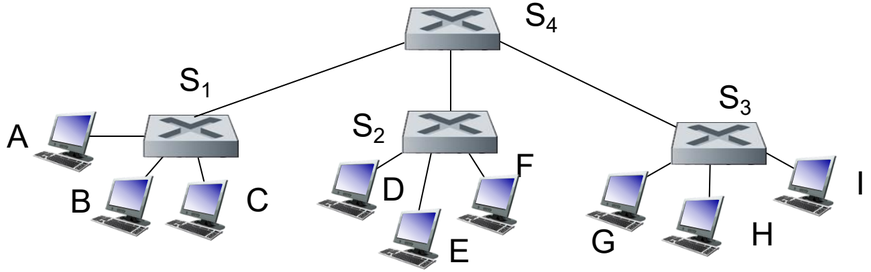
Suppose C sends frame to I, I responds to C



- Q: show switch tables and packet forwarding in S_1, S_2, S_3, S_4

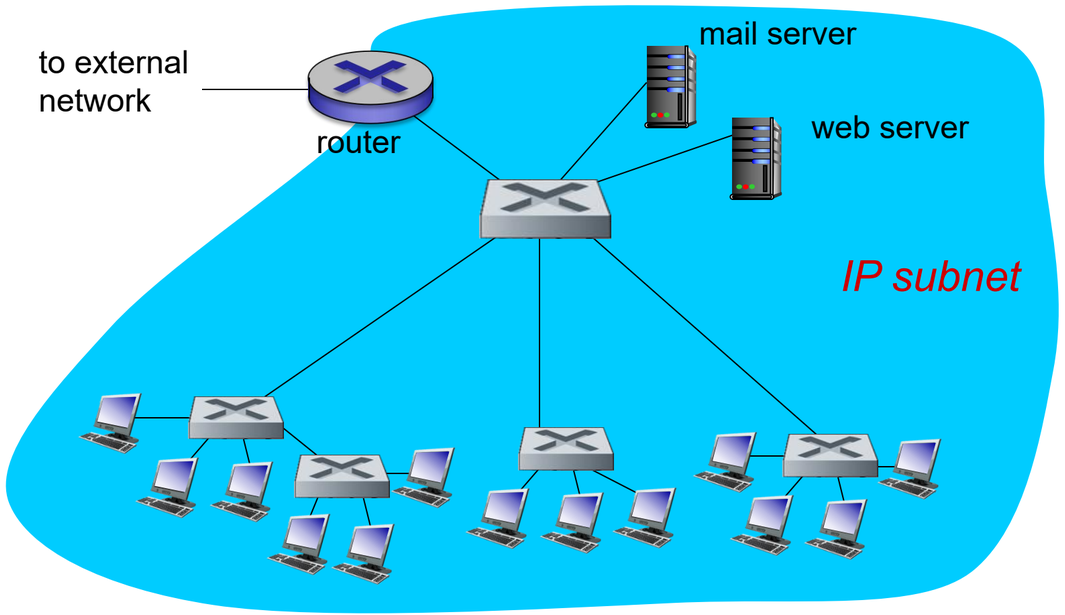
自学习多交换机示例

假设C向I发送帧，I向C响应

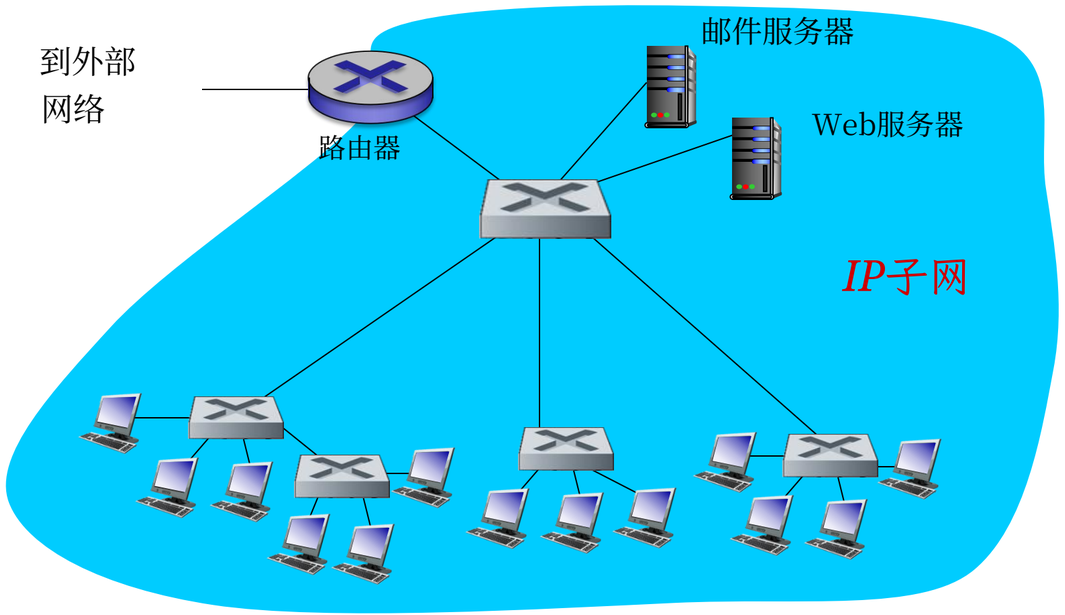


- Q: 在S中显示交换表和分组转发$_1$，S_2, S_3, S_4

Institutional network



机构网络



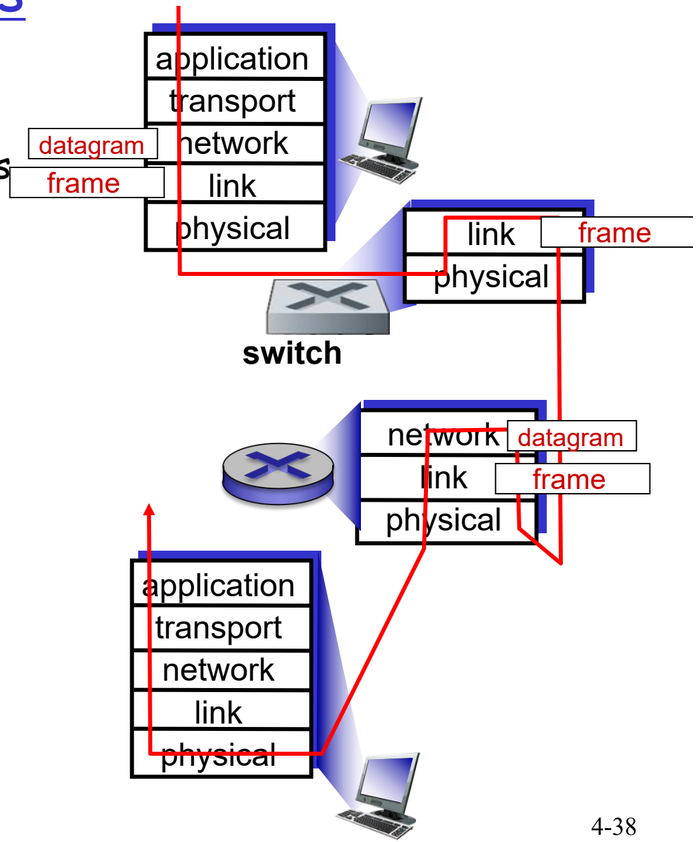
Switches vs. routers

both are store-and-forward:

- *routers:* network-layer devices (examine network-layer headers)
- *switches:* link-layer devices (examine link-layer headers)

both have forwarding tables:

- *routers:* compute tables using routing algorithms, IP addresses
- *switches:* learn forwarding table using flooding, learning, MAC addresses



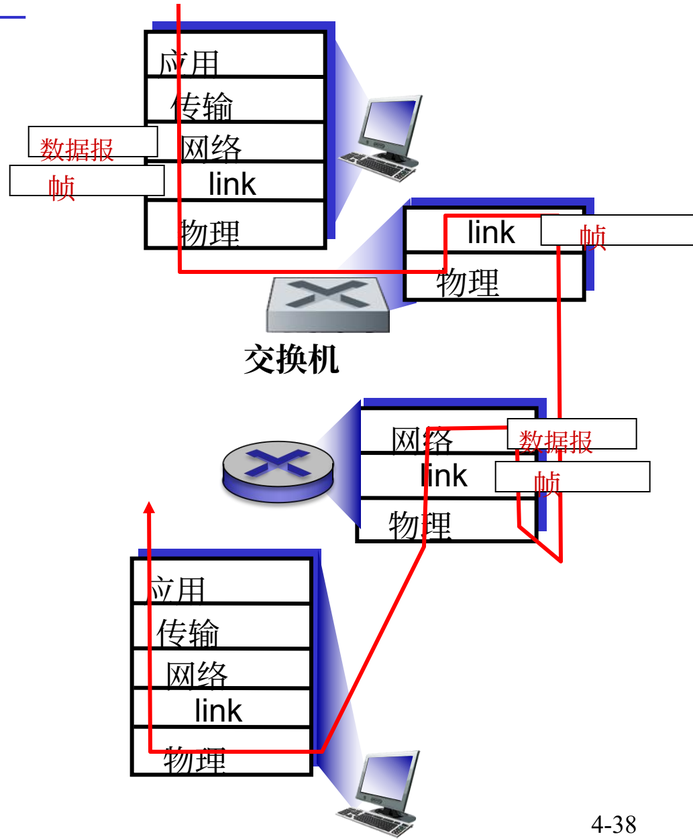
交换机与路由器

两者都采用存储转发方式：

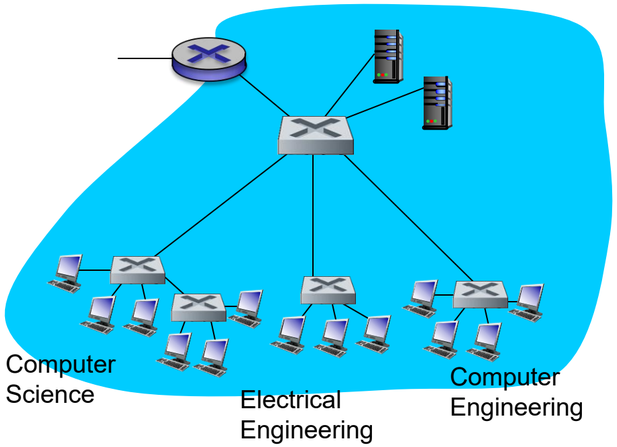
- 路由器：网络层设备 (检查网络层标题)
- 交换机：数据链路层设备 (检查数据链路层标题)

两者都有转发表：

- 路由器：使用路由算法计算表
路由算法，IP地址
- 交换机：使用泛洪和学习来学习MAC地址转发表



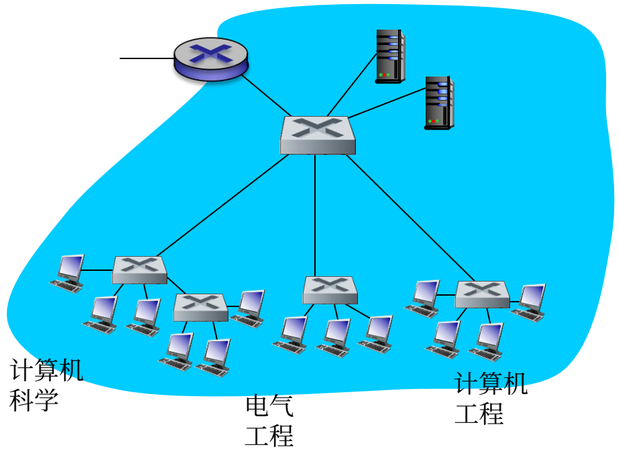
VLANs: motivation



consider:

- CS user moves office to EE, but wants connect to CS switch?
- single broadcast domain:
  - all layer-2 broadcast traffic (ARP, DHCP, unknown location of destination MAC address) must cross entire LAN
  - security/privacy, efficiency issues

虚拟局域网：动机



考虑：

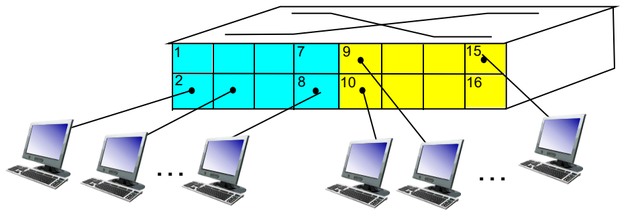
- CS用户将办公室搬到EE，但想连接到CS交换机?
- 单个广播域：
  - 所有二层广播流量（ARP、DHCP、未知位置目标MAC地址)必须跨越整个局域网
  - 安全/隐私，效率问题

VLANs

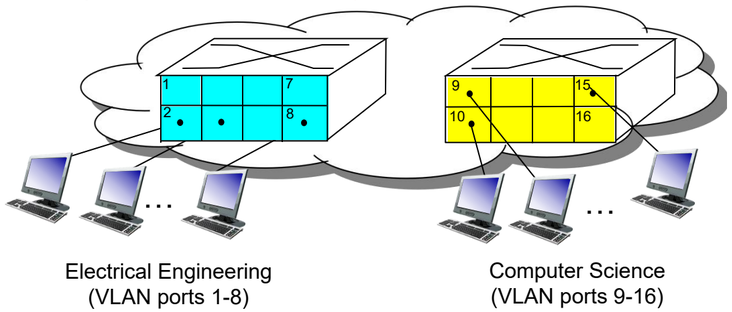
Virtual Local Area Network

switch(es) supporting VLAN capabilities can be configured to define multiple ***virtual*** LANS over single physical LAN infrastructure.

port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch ……



… operates as multiple virtual switches

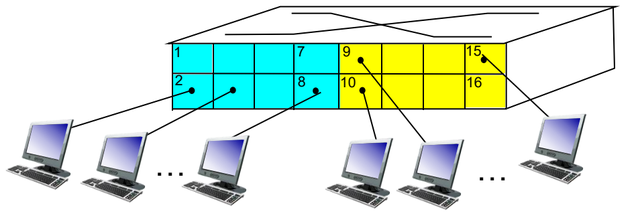


虚拟局域网

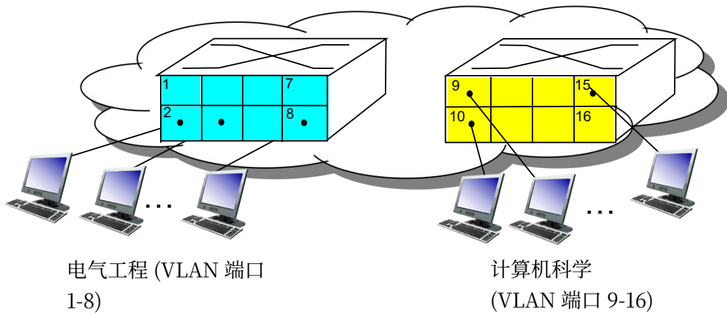
虚拟局域网

支持 VLAN 功能的交换机可以配置为在单个物理局域网基础设施上定义多个 虚拟 局域网。

基于端口的虚拟局域网： 交换机端口被（通过交换机管理软件）分组，以便 单个 物理交换机 …

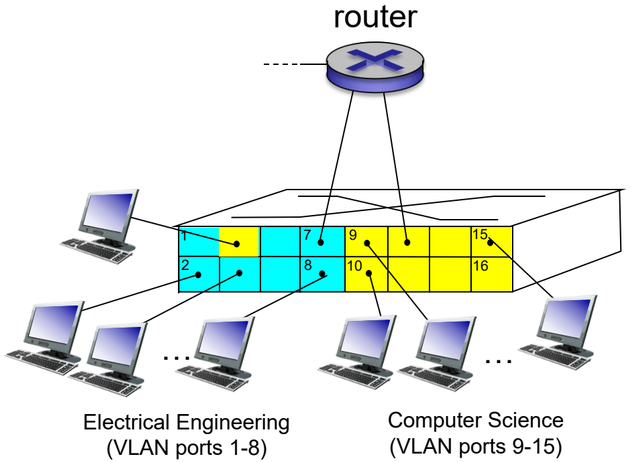


… 作为 多个 虚拟交换机



Port-based VLAN

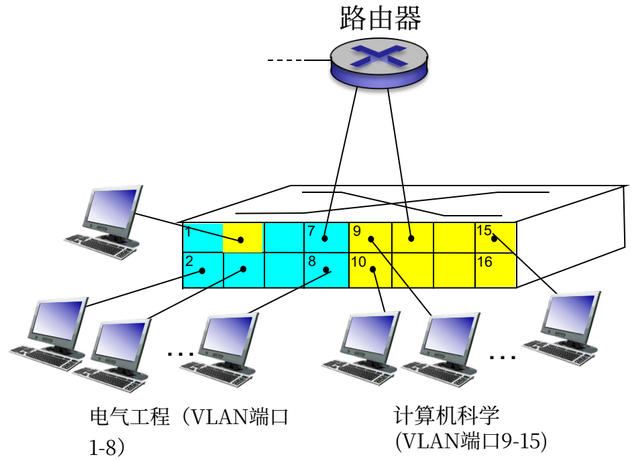
- *traffic isolation:* frames to/from ports 1-8 can *only* reach ports 1-8
  - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- dynamic membership: ports can be dynamically assigned among VLANs



- forwarding between VLANS: done via routing (just as with separate switches)
  - in practice vendors sell combined switches plus routers

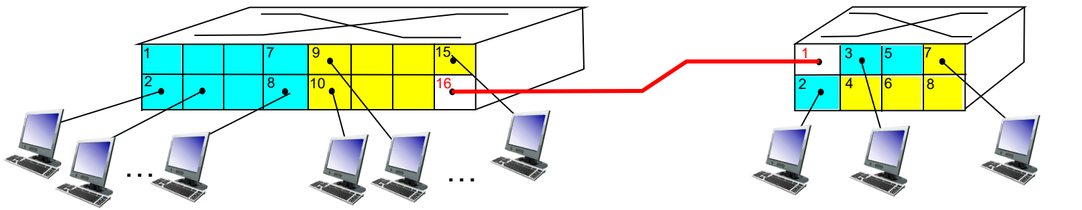
基于端口的虚拟局域网

- 流量隔离：端口1-8之间的帧可以仅访问端口1-8
  - 也可以根据端点的MAC地址定义VLAN，而不是交换机端口
- 动态成员：端口可以在虚拟局域网之间动态分配



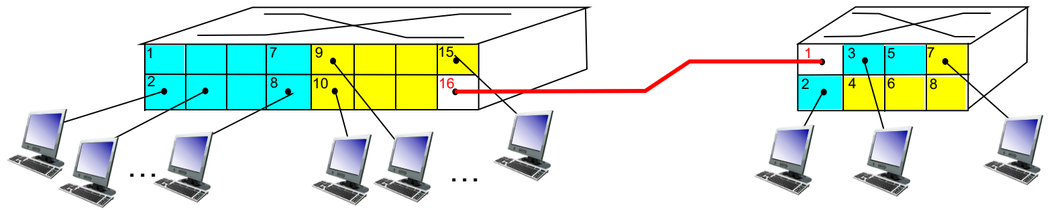
- 虚拟局域网之间的转发：通过路由完成（与独立交换机相同）
  - 实际上，供应商通常会销售集成的交换机加路由器

VLANS spanning multiple switches



- *trunk port:* carries frames between VLANS defined over multiple physical switches
  - frames forwarded within VLAN between switches can't be vanilla 802.1 frames (must carry VLAN ID info)
  - 802.1q protocol adds/removed additional header fields for frames forwarded between trunk ports

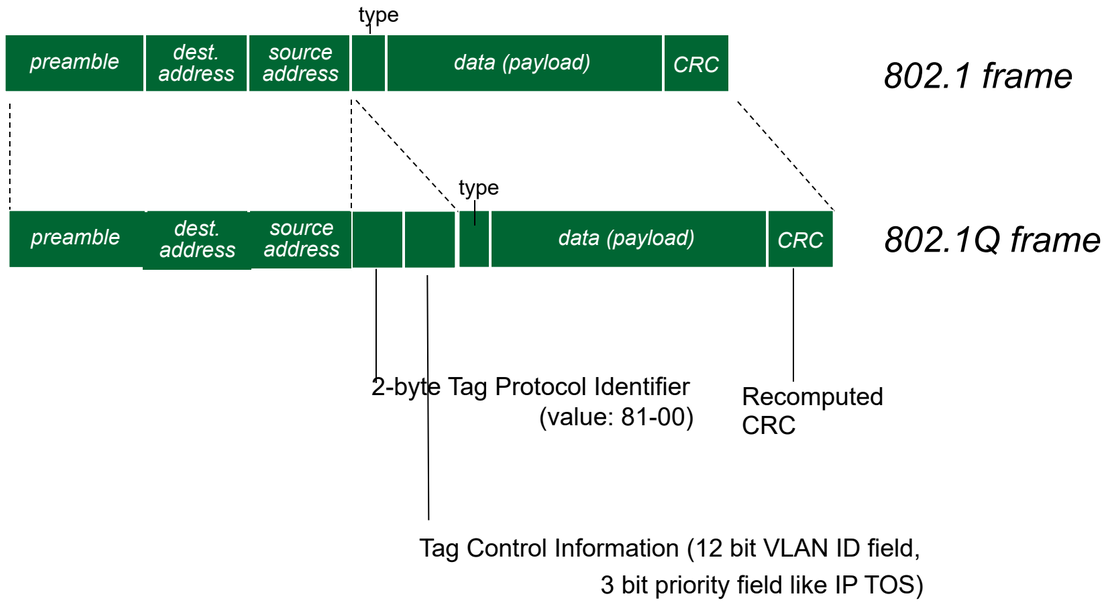
跨越多个交换机的虚拟局域网



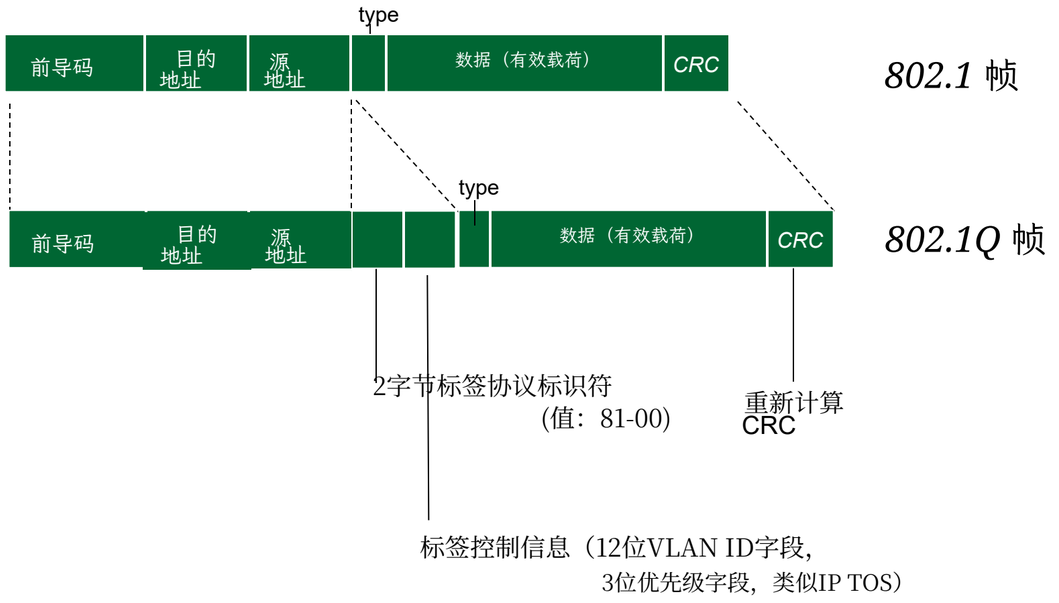
- *trunk*端口： 在多个物理交换机上定义的虚拟局域网之间传输帧
  - 在交换机之间转发VLAN内的帧时，不能是普通的802.1帧（必须携带VLAN ID信息）
  - 802.1q 协议为转发于中继端口之间的帧添加/移除附加的头部字段

中继端口

802.1Q VLAN frame format



802.1Q VLAN帧格式



Wireless communications

- the challenges of wireless communications
- wireless communication standards
- IEEE 802.11
- IEEE 802.15.4
- 5G?

无线通信

- 无线通信的挑战
- 无线通信标准
- IEEE 802.11
- IEEE 802.15.4
- 5G?

4-44

the challenges of Wireless communications

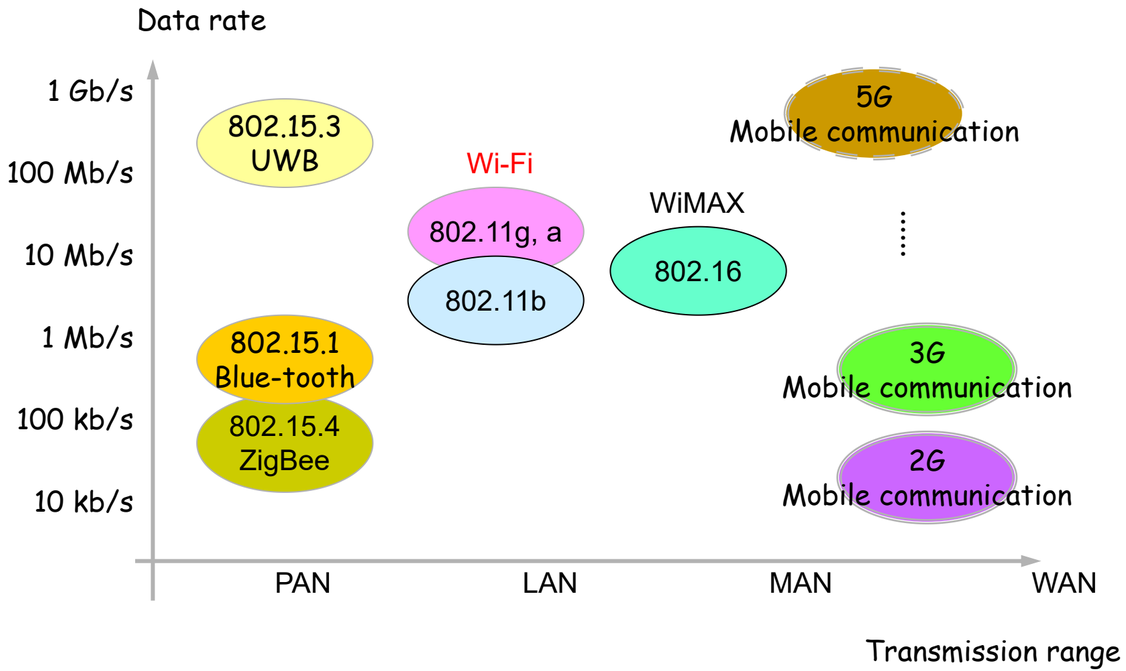
- fading: path loss, multipath effect, shadow effect, Doppler effect
- interference: from other wireless communications
- hidden terminal problem
- security
- mobility

无线通信的挑战

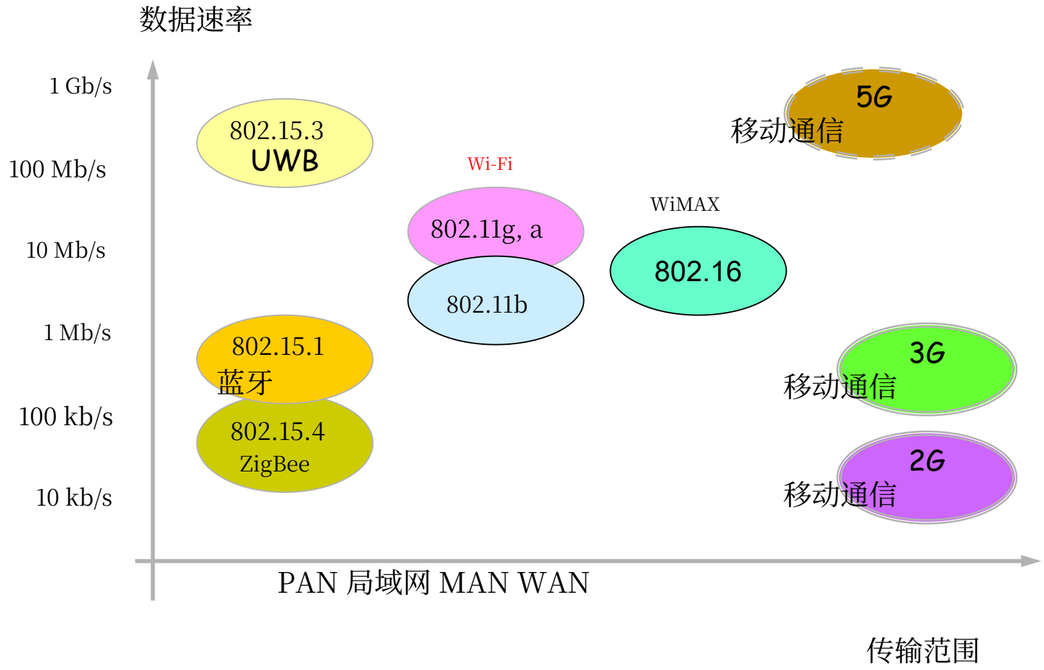
- 衰落：路径损耗、多径效应、阴影效应、多普勒效应
- 干扰：来自其他无线通信
- 隐藏终端问题
- 安全性
- 移动性

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Wireless communication technology



无线通信技术



Wireless communication technology

从一片空白到世界领先，中国通信翻身逆袭史_CSDN 程序人生的博客-CSDN博客

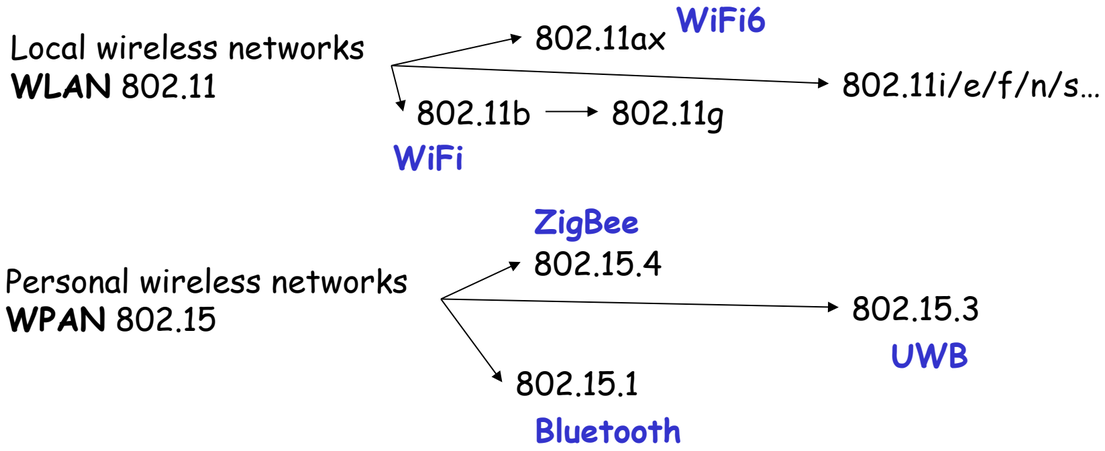
中国通讯发展史（六）1G到5G - 知乎 (zhihu.com)

无线通信技术

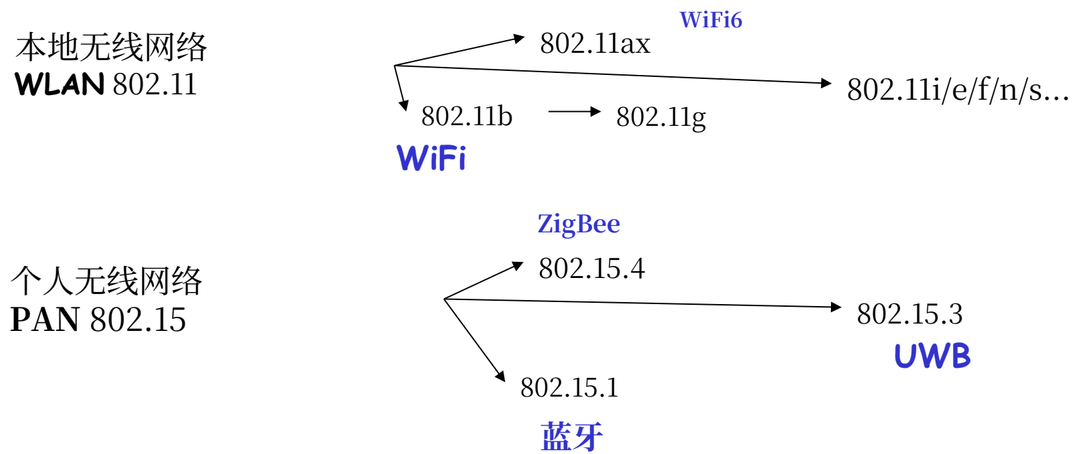
从一片空白到世界领先,中国通信翻身逆袭史_CSDN 程序人生的博客-CSDN博客

中国通讯发展史(六)1G到5G - 知乎 (zhihu.com)

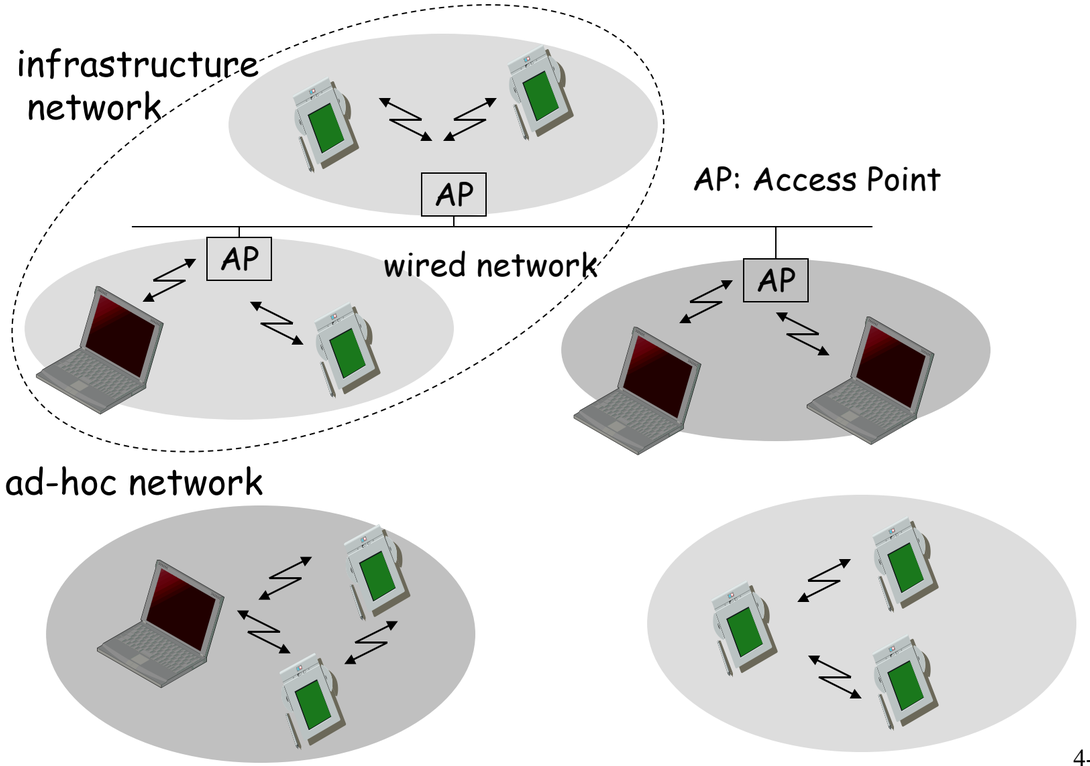
IEEE Wireless Technology



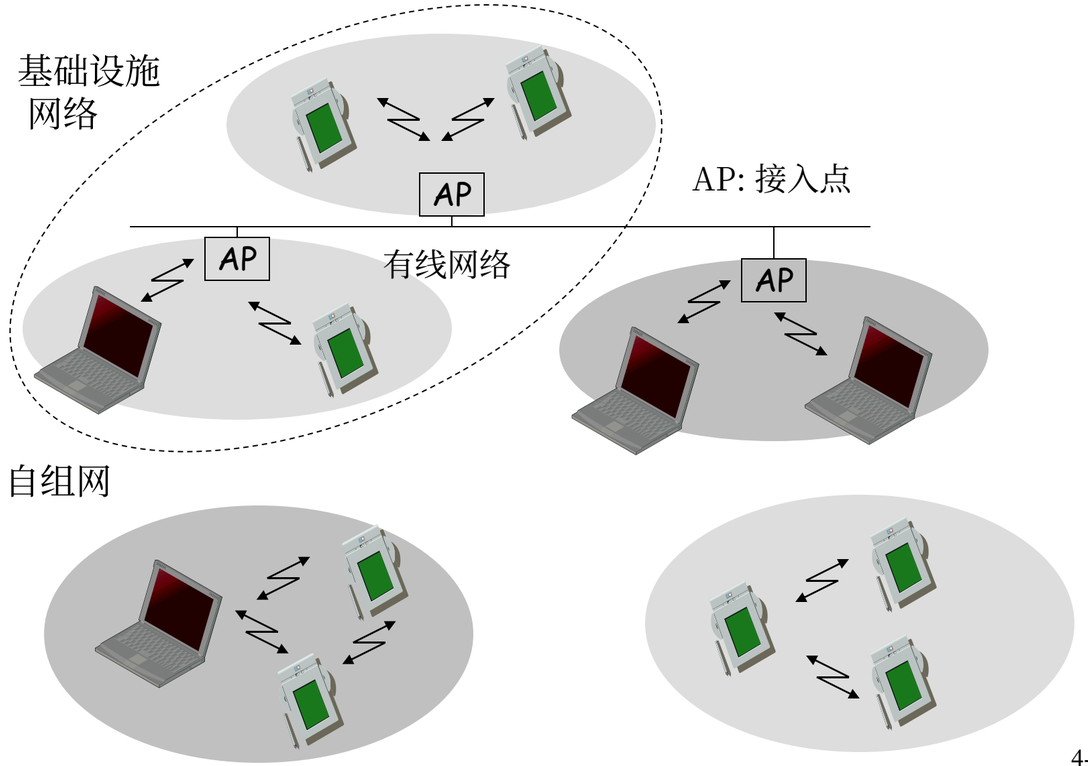
IEEE 无线技术



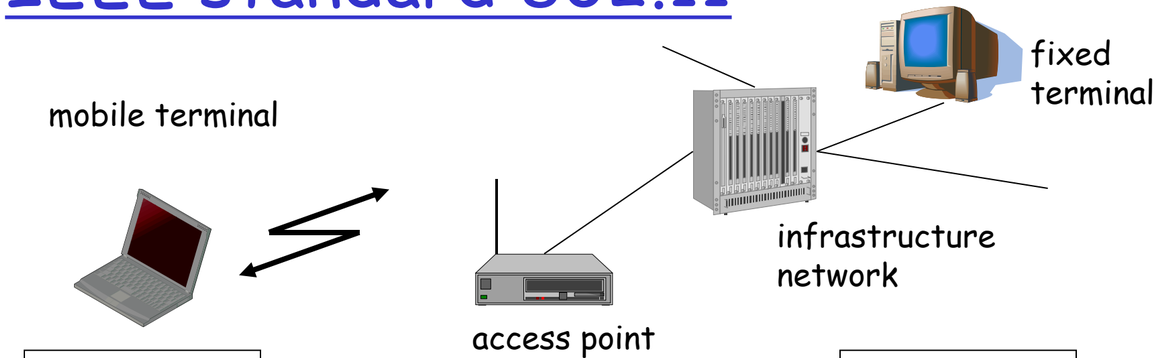
infrastructure vs. ad-hoc networks



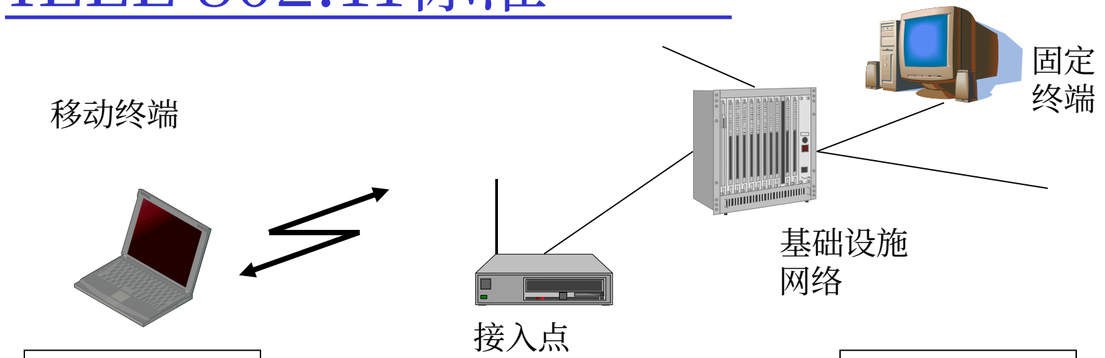
基础设施 vs. 自组网



IEEE standard 802.11



IEEE 802.11标准



IEEE 802.11 physical layer

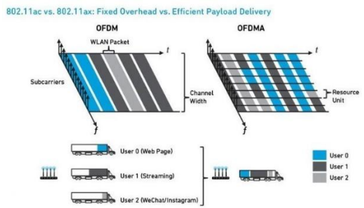
WiFi 版本	WiFi 标准	发布时间	最高速率	工作频段
WiFi 7	IEEE 802.11be	2022年	30Gbits	2.4GHz，5GHz，6GHz [4]
WiFi 6	IEEE 802.11ax	2019 年	11Gbps	2.4GHz 或 5GHz
WiFi 5	IEEE 802.11ac	2014 年	1Gbps	5GHz
WiFi 4	IEEE 802.11n	2009 年	600Mbps	2.4GhHz 或 5GHz
WiFi 3	IEEE 802.11g	2003 年	54Mbps	2.4GHz
WiFi 2	IEEE 802.11b	1999 年	11Mbps	2.4GHz
WiFi 1	IEEE 802.11a	1999 年	54Mbps	5GHz
WiFi 0	IEEE 802.11	1997 年	2Mbps	2.4GHz

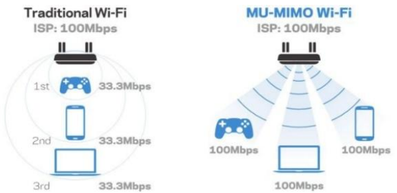
2.4GHz（802.11b/g/n/ax），5GHz（802.11a/n/ac/ax）

802.11协议发展历史

WiFi 6

- OFDMA
- MU-MIMO
- 1024-QAM
- Spatial Reuse & BBS Coloring





IEEE 802.11 物理层

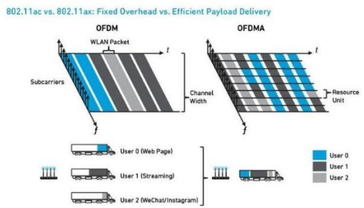
WiFi 版本	WiFi 标准	发布时间	最高速率	工作频段
WiFi 7	IEEE 802.11be	2022年	30Gbits	2.4GHz，5GHz，6GHz [4]
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WiFi 2	IEEE 802.11b	1999 年	11Mbps	2.4GHz
WiFi 1	IEEE 802.11a	1999 年	54Mbps	5GHz
WiFi 0	IEEE 802.11	1997 年	2Mbps	2.4GHz

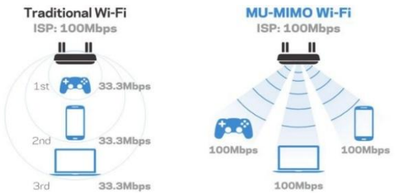
2.4GHz（802.11b/g/n/ax），5GHz（802.11a/n/ac/ax）

802.11 协议发展历史

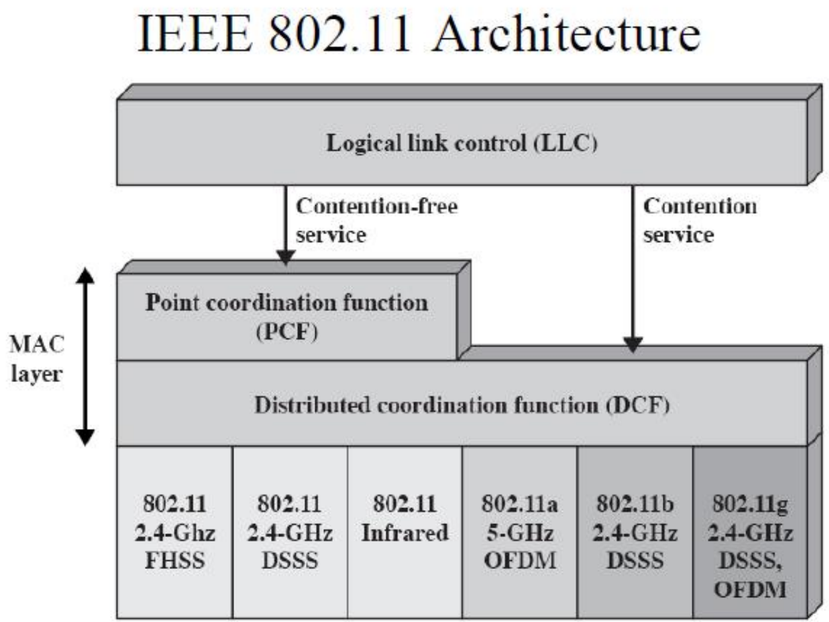
WiFi 6

- OFDMA
- MU-MIMO
- 1024-QAM
- 空间复用 & BBS 调色



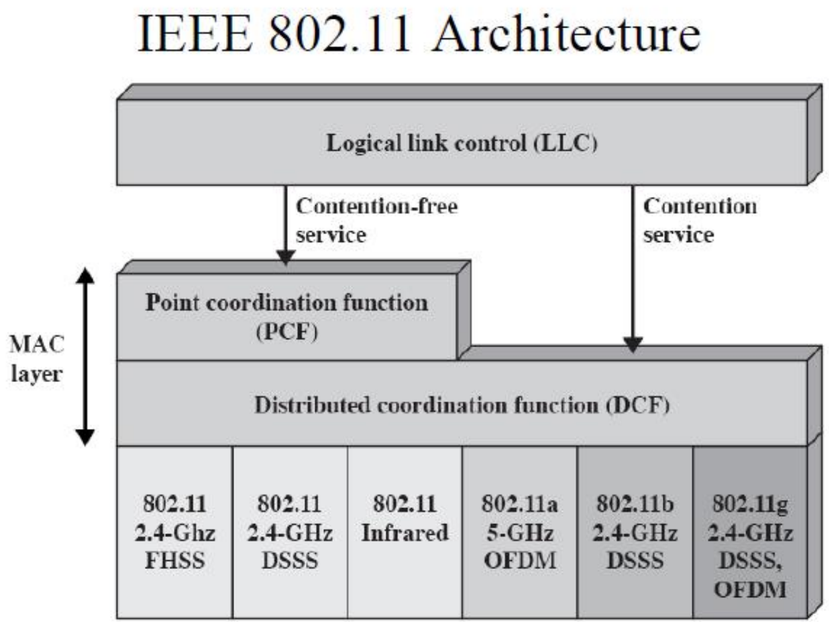


IEEE 802.11 protocol architecture



IEEE 802.11 Protocol Architecture

IEEE 802.11 协议架构



IEEE 802.11 Protocol Architecture

802.11 - MAC layer

- Traffic services
  - Asynchronous Data Service (mandatory)
    - implemented using DCF (Distributed Coordination Function)
  - Time-Bounded Service (optional)
    - implemented using PCF (Point Coordination Function)
- Access methods
  - DCF CSMA/CA (mandatory)
    - Distributed Wireless MAC
    - collision avoidance via randomized „back-off" mechanism
    - minimum distance between consecutive packets
    - ACK packet for acknowledgements (not for broadcasts)
  - DCF w/ RTS/CTS (optional)
    - avoids hidden terminal problem
  - PCF (optional)
    - access point polls terminals
    - Contention free

802.11 - MAC层

- 流量服务
  - 异步数据服务（强制）
    - 使用DCF（分布式协调功能）实现 on)
  - 时间限制服务（可选）
    - 使用PCF（点协调功能）实现
- 访问方法
  - DCF CSMA/CA (强制)
    - 分布式无线MAC
    - 冲突避免 通过随机„退避“机制
    - 连续数据包之间的最小间隔
    - ACK数据包用于确认（不用于广播）
  - DCF w/ RTS/CTS（可选）
    - 避免隐藏终端问题
  - PCF（可选）
    - 接入点轮询终端
    - 无竞争

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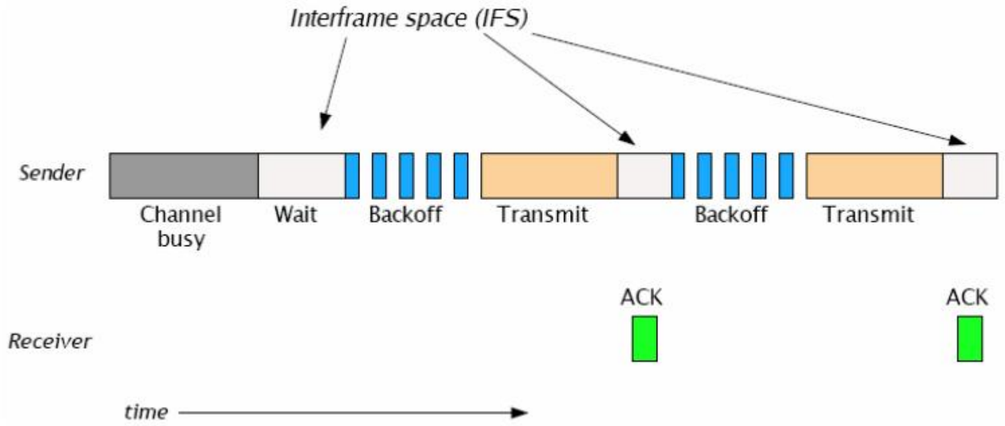
802.11 MAC functions

- MAC layer covers three functional areas:
  - Reliable data delivery
    - ACK-based scheme for reliability (receiver sends ACK after each successful transmission)
  - Medium access control
    - CSMA/CA; collision avoidance, not collision detection, Why? How?
  - Security
    - Wired Equivalent Privacy (WEP),WEP relies on a secret key being shared by end hosts and APs

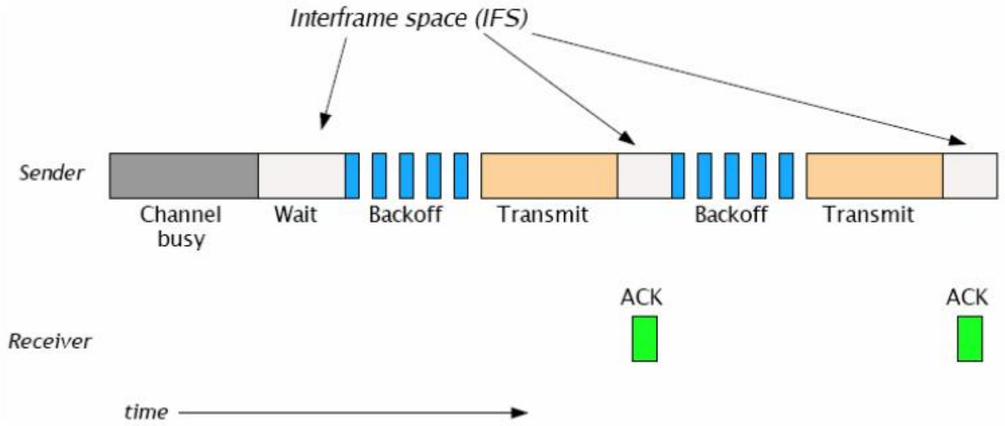
802.11 介质访问控制功能

- MAC层涵盖三个功能区域:
  - 可靠数据传输
    - 基于ACK的方案 用于可靠性（接收器在每次成功传输后发送ACK）
  - 介质访问控制
    - CSMA/CA；冲突避免而非冲突检测， 为什么? 如何?
  - 安全性
    - 有线等效保密 (WEP)， WEP 依赖于终端主机和接入点共享的密钥

DCF CSMA/CA Illustrated

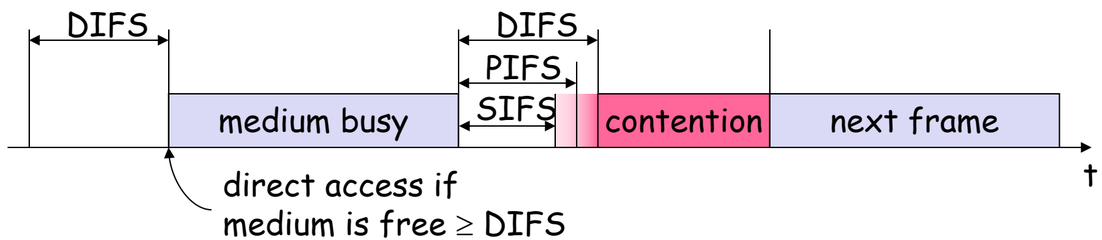


DCF CSMA/CA Illustrated



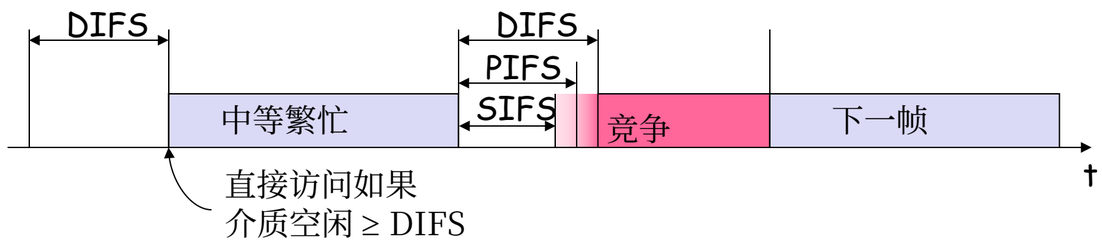
802.11 - MAC

- Priorities
  - defined through different inter frame spaces
  - SIFS (Short Inter Frame Spacing) :
    - 10µs (802.11b/g), 16 µs (802.11a)
    - High priority, for ACK, CTS, polling response
  - PIFS (PCF IFS) :
    - PIFS = SIFS + Slot time, which is 20 µs 802.11b, 9 µs 802.11a/g
    - medium priority, for time-bounded service using PCF
  - DIFS (DCF IFS):
    - DIFS = PIFS + Slot time
    - lowest priority, for asynchronous data service

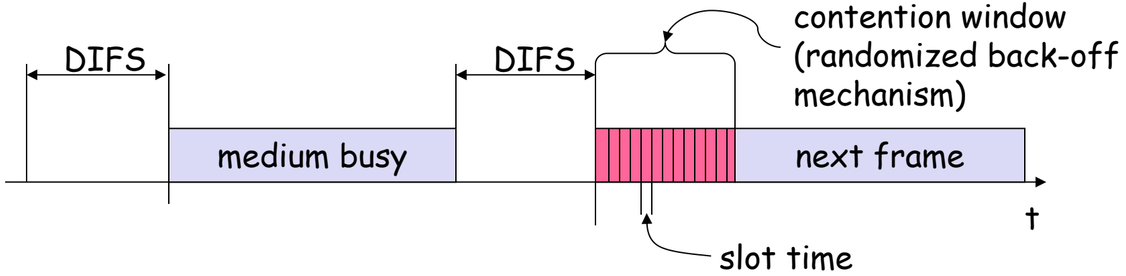


802.11 - MAC

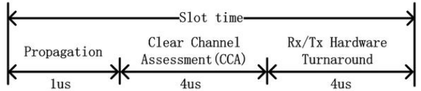
- 优先级
  - 通过不同的帧间空间定义
  - SIFS（短帧间空间）：
    - 10µs (802.11b/g), 16 µs (802.11a)
    - 高优先级，用于ACK、CTS、轮询响应
  - PIFS（PCF IFS）：
    - PIFS = SIFS + 时隙时间，对于802.11b是20 µs，802.11a/g是9 µs
    - 中等优先级，用于使用PCF的时间受限服务
  - DIFS (DCF IFS)：
    - DIFS = PIFS + 时隙时间
    - 最低优先级，用于异步数据服务



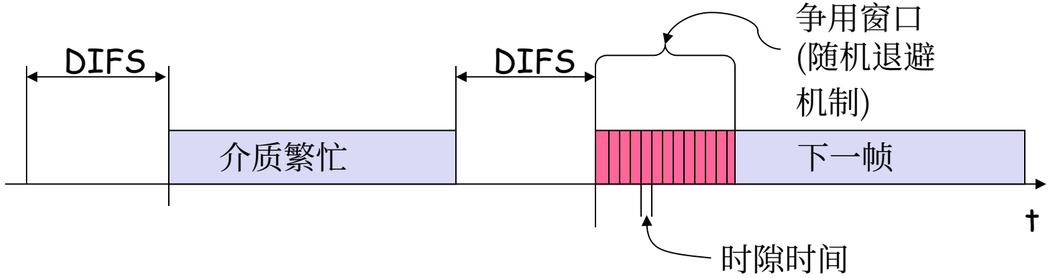
CSMA/CA access method



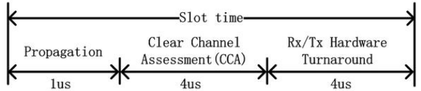
- station ready to send starts sensing the medium (Carrier Sense based on CCA, Clear Channel Assessment)
- if the medium is free for the duration of an Inter-Frame Space (IFS), the station can start sending (IFS depends on service type)
- if the medium is busy, the station has to wait for a free IFS, then the station must additionally wait a random back-off time (collision avoidance, multiple of slot-time)
  - Slot time = 20 µs for 802.11b, 9 µs in 802.11a/g
  - CW_min = 16 for 802.11a. 32 for 802.11b
  - CW_max = 1024



CSMA/CA 访问方法



- 站点准备发送时开始感知信道（基于载波侦听CCA的信道感知）
- 如果介质在帧间空间（IFS）的持续时间内是空闲的，站点可以开始发送（IFS取决于服务类型）
- 如果介质是繁忙的，站点必须等待空闲的IFS，然后站点还必须额外等待一个随机退避时间（冲突避免，时隙时间的倍数）
  - 时隙时间 = 20 µs 用于 802.11b，9 µs 用于 802.11a/g
  - CW_最小 = 16 用于 802.11a，32 用于 802.11b
  - CW_max = 1024



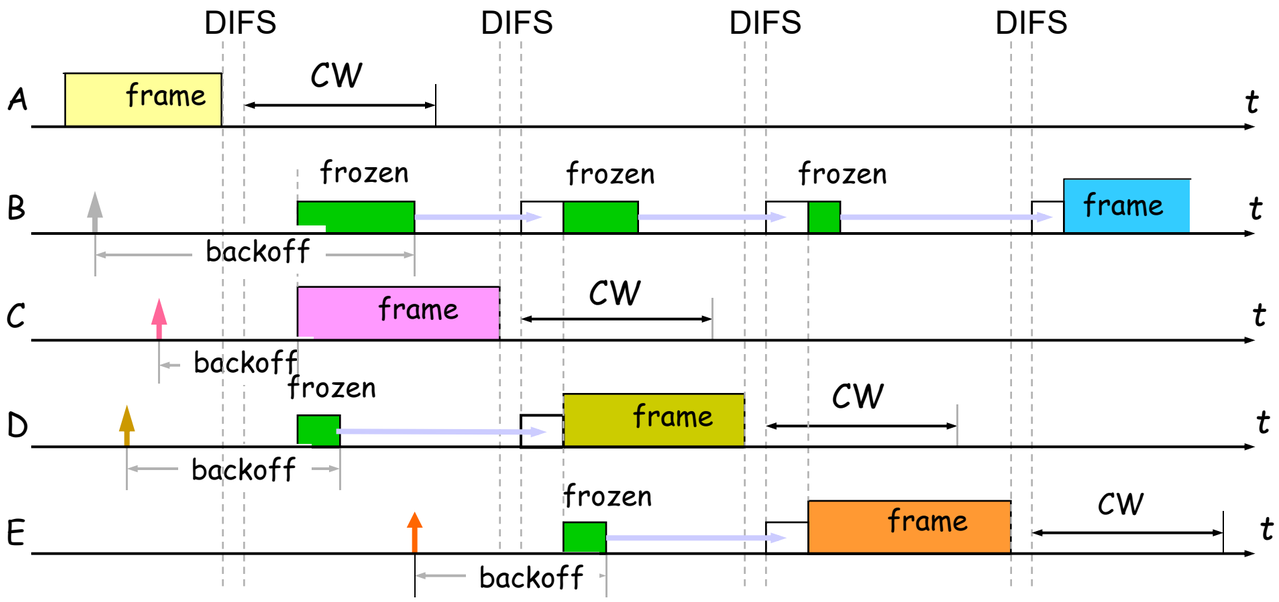
CSMA/CA access method (Continued)

- If another station occupies the medium during the back-off time of the station, the back-off timer stops (fairness)
- When back-off timer reaches zero, start transmission
  - If more than one nodes decrement to zero at the same time, a collision will occur.
- If a collision occurs (missing ACK), the corresponding nodes double the CW size and choose their back-off time from the increase CW
- After successful transmission, CW size is reset to its min value.

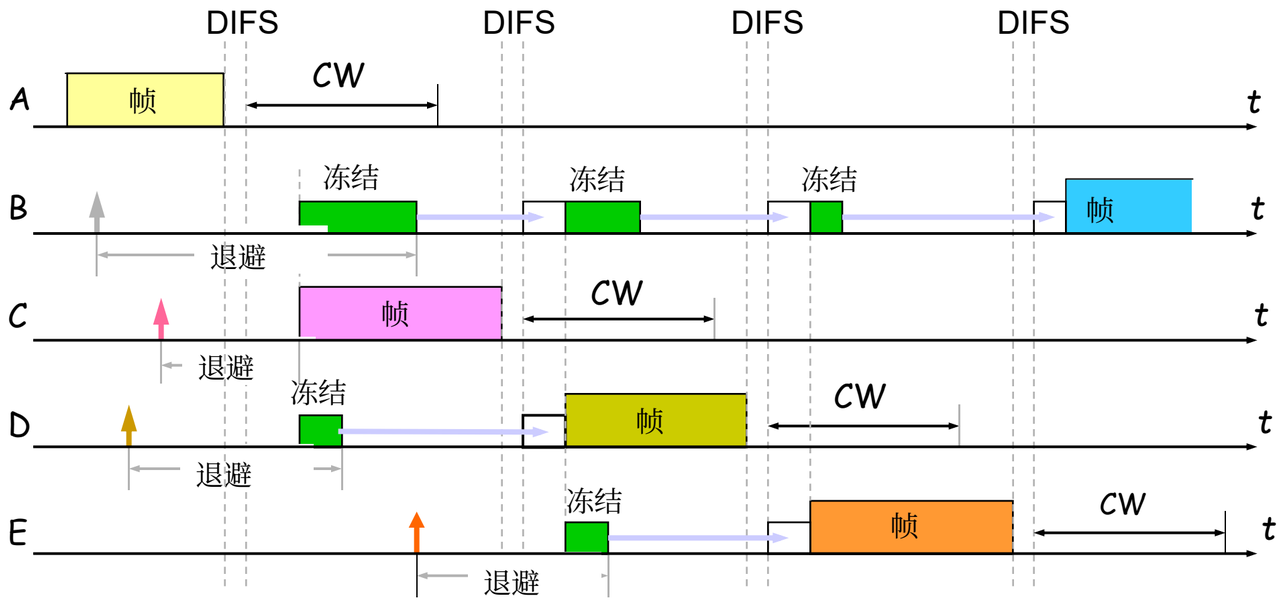
CSMA/CA 访问方法（续）

- 如果另一个站点在站点的退避时间内占用了介质，退避定时器停止（公平性）
- 当退避计时器达到零时，开始传输
  - 如果多个节点同时减到零，就会发生冲突。
- 如果发生冲突（ACK丢失），相应的节点 将CW大小加倍 并从增加的CW中选择它们的退避时间
- 传输成功后，CW大小重置为其最小值。

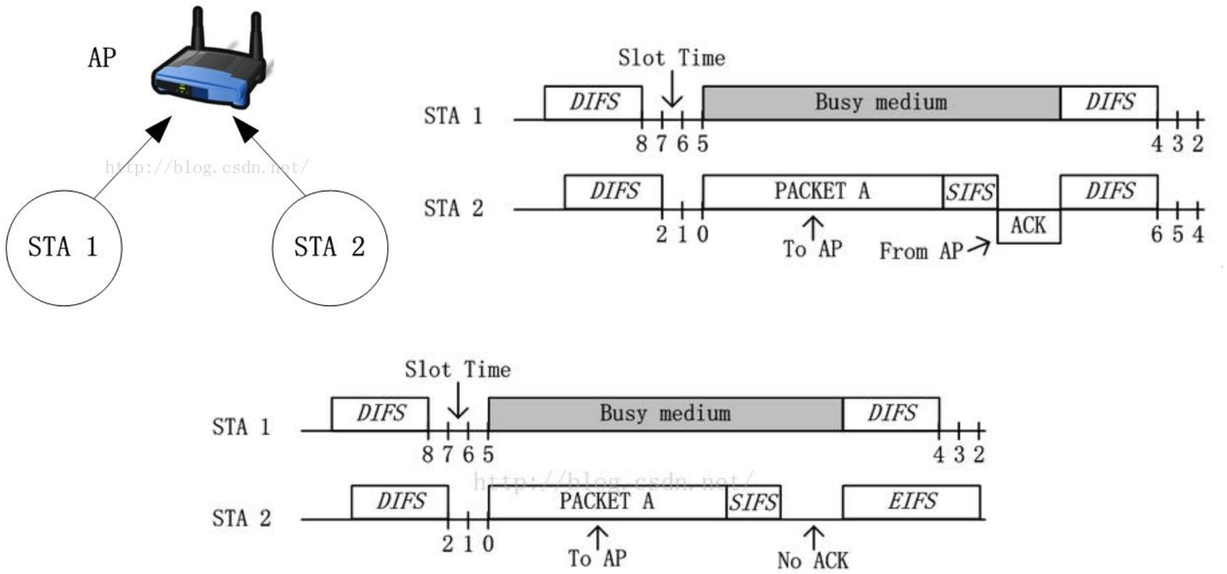
A simplified example



一个简化的示例



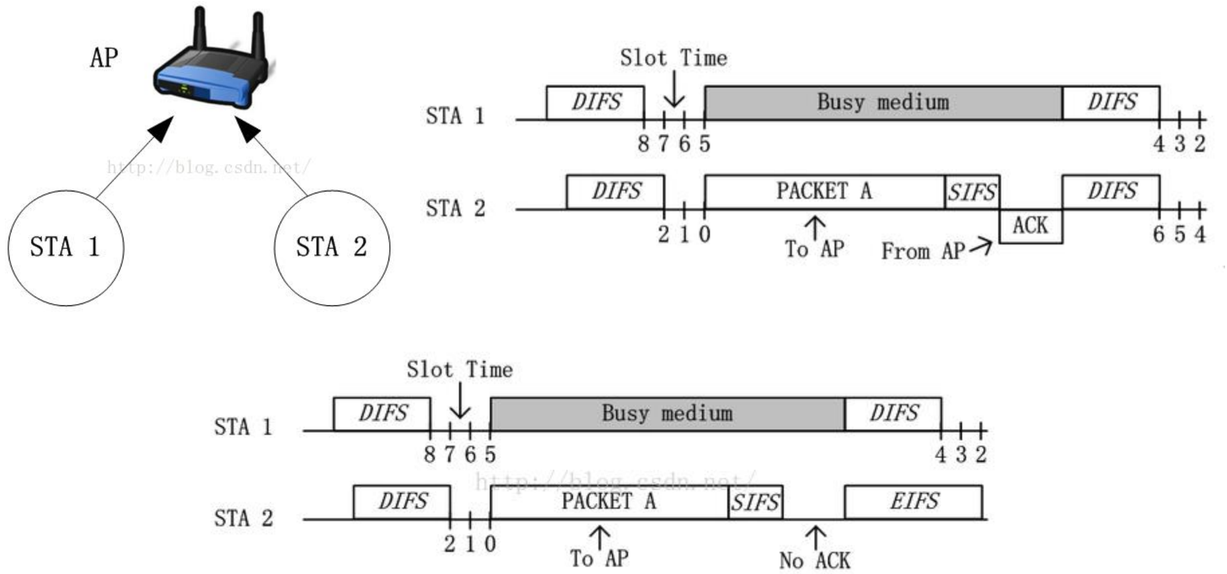
CSMA/CA example



802.11协议精读2：DCF与CSMA/CA_adr5970的博客-CSDN博客

https://www.zhihu.com/column/dot11

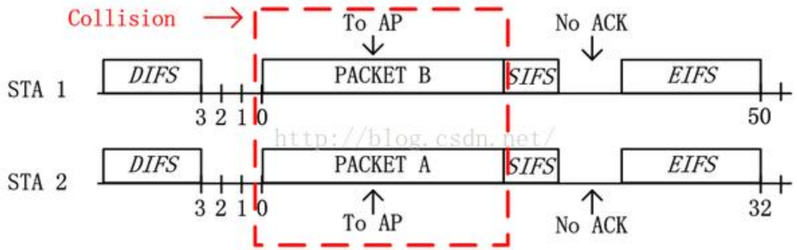
CSMA/CA 示例



802.11协议精读2:DCF与CSMA/CA_adr5970的博客-CSDN博客

https://www.zhihu.com/column/dot11

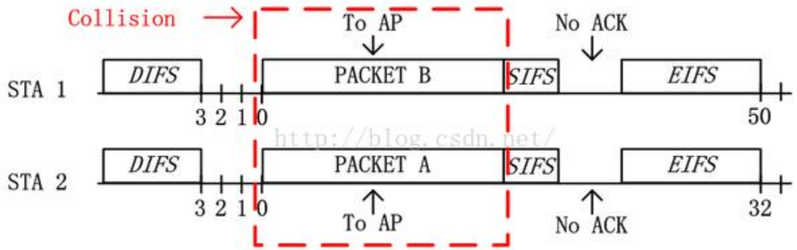
CSMA/CA example



在 **"等待"** DIFS后，STA 1与STA 2从各自的竞争窗体CW中选择一个随机数。只是碰巧的是，两者随机到了一样的数值，如图中，STA 1与STA 2都是随机到了3作为随机回退计数值。在经过3个slot time之后，因为两者同一时候倒数至0。那么意味着两者会同一时候发送数据。如图中的红色虚线框表示，在AP处因为两者信号互相干扰，从而都无法正确解码，从而CRC校验错误。即发生冲突。在冲突之后，即若AP处CRC校验失败，则不会给随意节点反馈ACK数据包。故两节点在ACK timeout之后，则等待EIFS之后，准备进入下一次竞争。

802.11协议精读2：DCF与CSMA/CA_adr5970的博客-CSDN博客

CSMA/CA示例



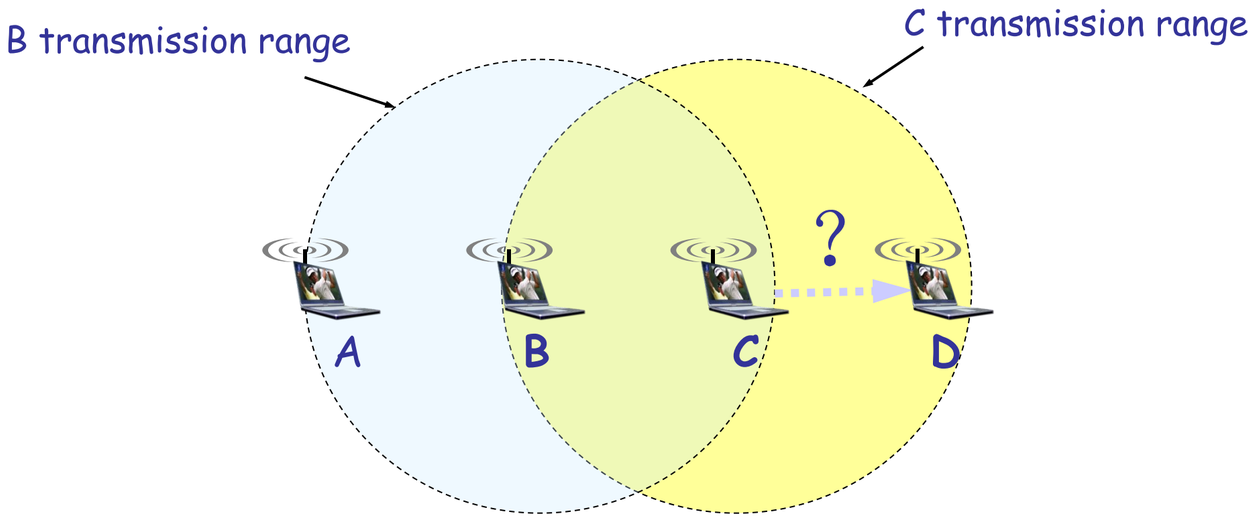
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C

校验失败，则不会给随意节点反馈ACK数据包。故两节点在ACK超时之后，则等待EIFS之后，准备进入下一次竞争。

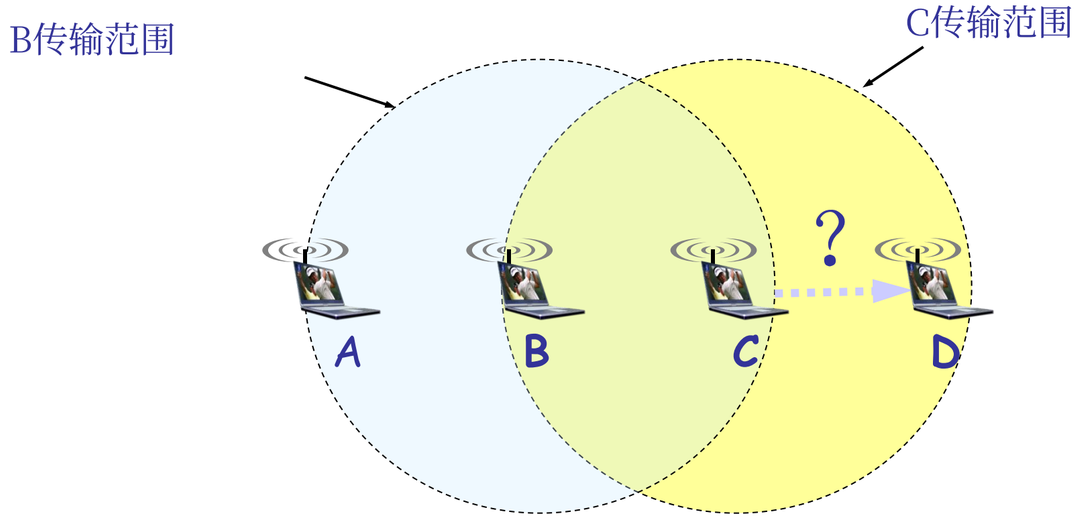
802.11协议精读2:DCF与CSMA/CA_adr5970的博客-CSDN博客

LAN exposed terminal problem



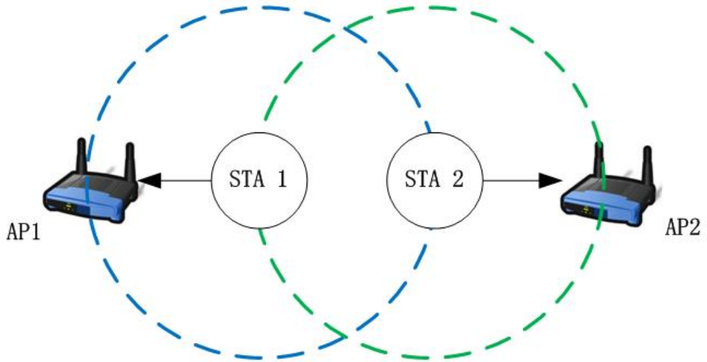
B is sending data to A. At the same time, C hopes to communication with D. But C senses a signal in the medium and dare not send the data.

局域网暴露终端问题



B 向 A 发送数据。同时，C 准备与 D 进行通信。但 C 感觉到介质中有信号，不敢发送数据。

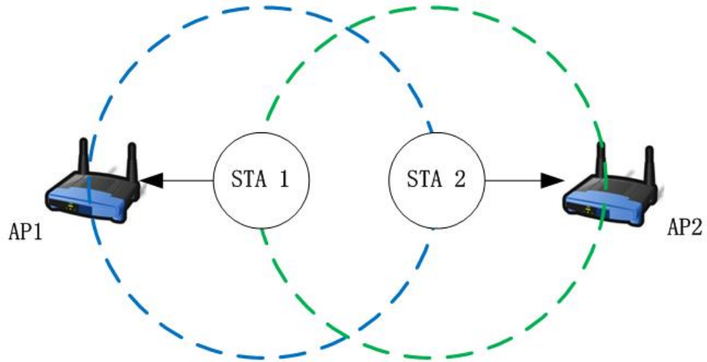
LAN exposed terminal problem



AP1处于STA 1的覆盖范围内，而不再STA 2的覆盖范围内。AP2处于STA 2的覆盖范围，而不在STA 1的覆盖范围内。换言之，AP1只能接受到STA 1的数据，AP2也只能接收到STA 2的数据。当STA 1与STA 2同时发送时，接受节点AP1或者AP2处均不会发生冲突，故其是可以同时传输的。但是由于这样的拓扑特殊性以及DCF中CSMA/CA的工作机制，造成STA 1与STA 2无法同时传输，该问题则是暴露终端问题。

802.11协议精读5：隐藏终端和暴露终端 - 知乎 (zhihu.com)

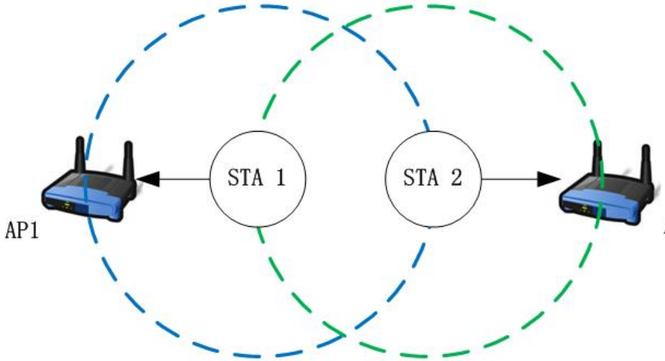
局域网暴露终端问题



AP1处于STA 1的覆盖范围内，而不再STA 2的覆盖范围内。AP2处于STA 2的覆盖范围，而不在STA 1的覆盖范围内。换言之，AP1只能接受到STA 1的数据，AP2也只能接收到STA 2的数据。当STA 1与STA 2同时发送时，接受节点AP1或者AP2处均不会发生冲突，故其是可以同时传输的。但是由于这样的拓扑特殊性以及DCF中CSMA/CA的工作机制，造成STA 1与STA 2无法同时传输，该问题则是暴露终端问题。

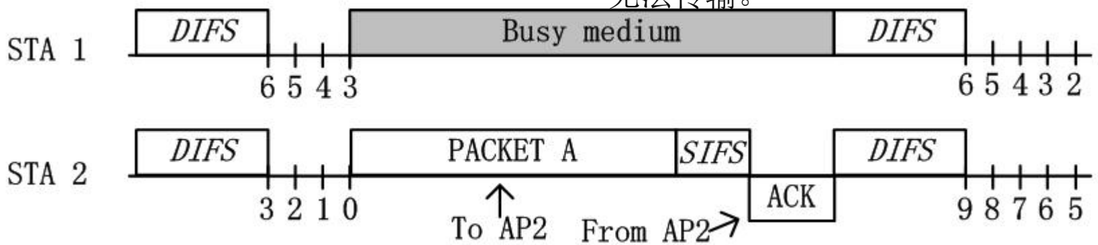
802.11协议精读5:隐藏终端和暴露终端 - 知乎 (zhihu.com)

LAN exposed terminal problem



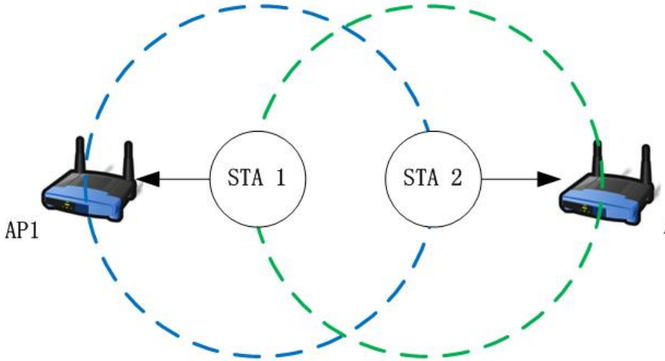
物理载波监听引起的暴露终端

由于STA 1与STA 2可以互相监听。由于STA 2选择了较小的随机数进行倒数，从而其最先倒数至0，并进行发送。当STA 2首先发送数据包给STA 2后，STA 1监听信道为忙状态，从而无法发送信息。故根据拓扑而言，STA 1是可以传数据给AP1的，但是由于监听STA 2正在传输，导致信道忙，故STA1悬挂随机倒数计数器，无法继续倒数，从而无法传输。



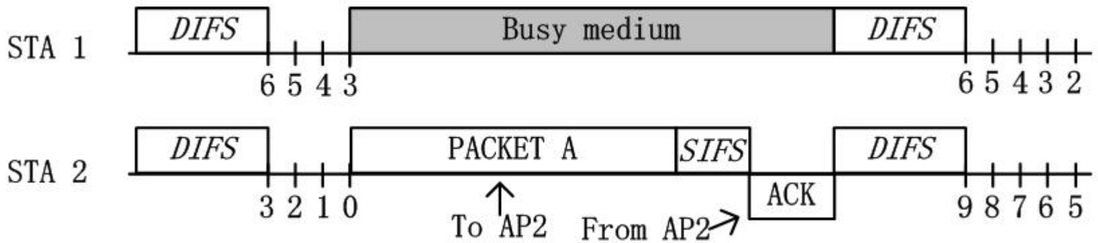
802.11协议精读5：隐藏终端和暴露终端 - 知乎 (zhihu.com)

局域网暴露终端问题



物理载波监听引起的暴露终端

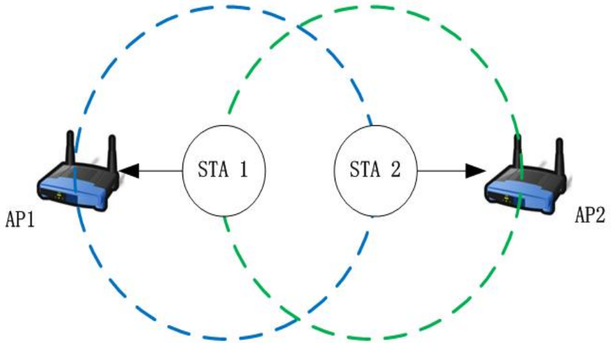
由于STA 1与STA 2可以互相监听。由于STA 2选择了较小的随机数进行倒数 ,从而其最先倒数至0,并进行发送。当 STA 2首先发送数据包给STA 2后, STA 1监听信道为忙状态,从而无法发 送信息。故根据拓扑而言, STA 1是可 以传数据给AP1的,但是由于监听 STA 2正在传输,导致信道忙,故STA1悬挂 随机倒数计数器,无法继续倒数,从而 无法传输。



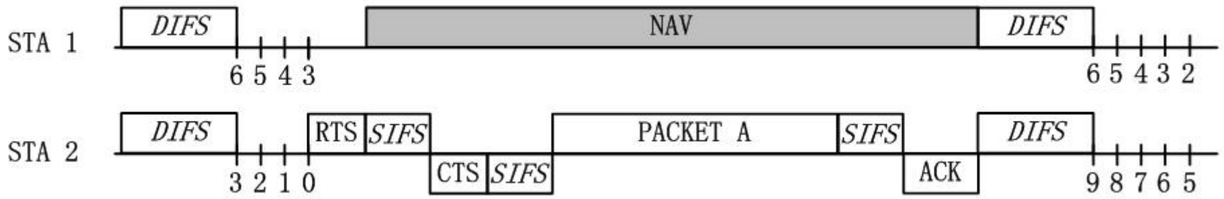
802.11协议精读5:隐藏终端和暴露终端 - 知乎 (zhihu.com)

LAN exposed terminal problem

虚拟载波监听引起的暴露终端



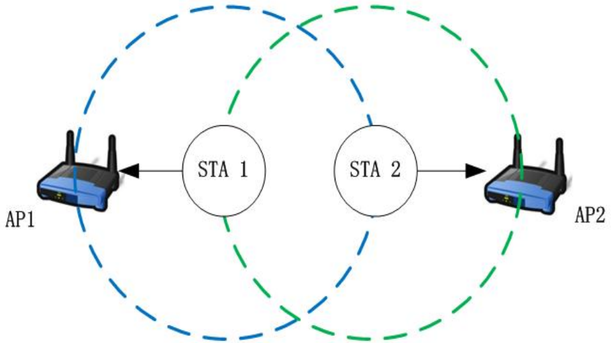
在暴露终端场景中，若STA 2不仅选择了较小的随机数进行优先倒数，并且其发送的数据包是RTS数据包。当STA 1识别到该RTS数据包后，其就会被设置为NAV状态，无法在后面的过程主动竞争信道，进而无法传输。与之前描述用RTS/CTS解决隐藏终端问题时不同，在解决隐藏终端问题中，NAV是由AP所反馈的CTS帧所进行保护。而这里由于STA 1与STA 2能够互相监听，换言之，在暴露终端情况下，STA 1的NAV是被STA 2所发送的RTS帧进行保护的。在STA 1被NAV保护后，其也无法传输，最终导致暴露终端问题。



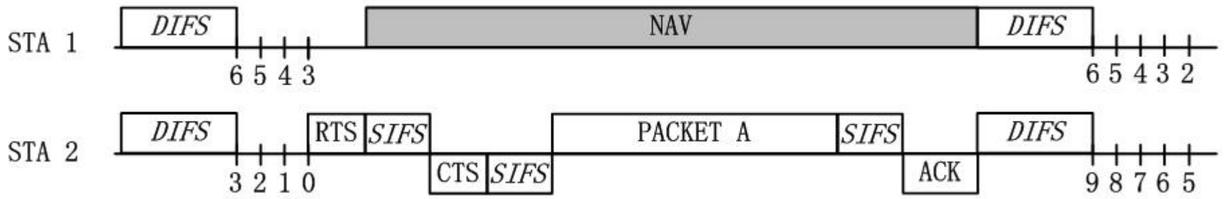
802.11协议精读5：隐藏终端和暴露终端 - 知乎 (zhihu.com)

局域网暴露终端问题

虚拟载波监听引起的暴露终端

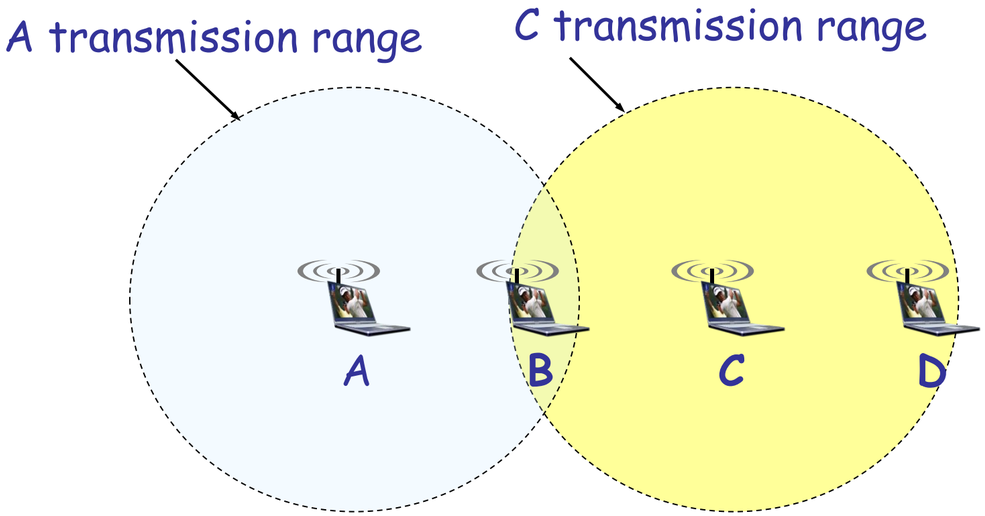


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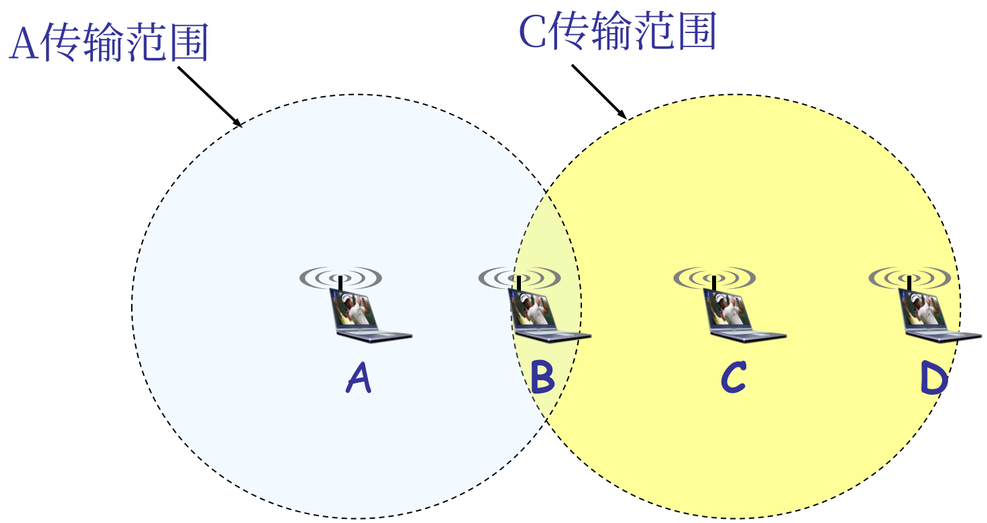
802.11协议精读5:隐藏终端和暴露终端 - 知乎 (zhihu.com)

LAN hidden terminal problem



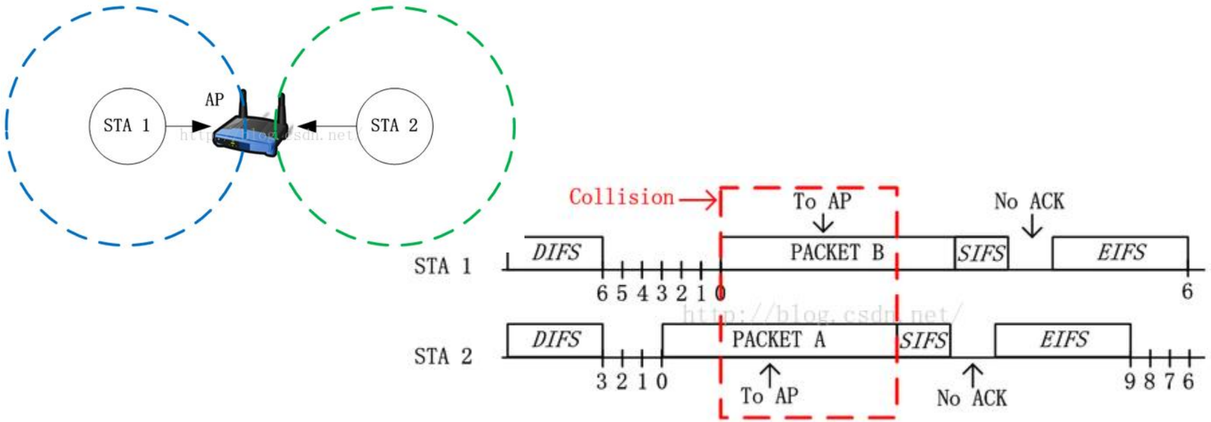
A and C cannot hear each other and think B is idle, then, they both send data to B. A collision appears at the destination, B.

LAN隐藏终端问题



A和C互相听不到，认为B处于空闲状态，然后，他们两人都向B发送数据。在目的地B处出现碰撞。

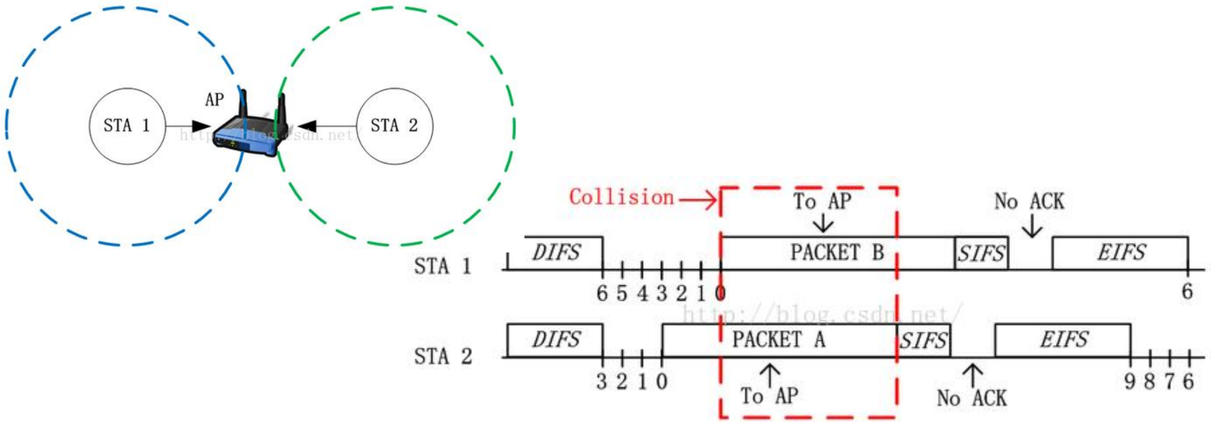
LAN hidden terminal problem



因为STA 1与STA 2无法互相监听，即STA 2发送数据后，STA 1还继续进行backoff过程，从而继续倒数。当STA 1的随机回退计数值倒数至0时。STA 1也会发送数据。因为STA 1与STA 2的发送存在重叠区域，即也是发生了冲突，AP无法正确接收数据。即不会反馈ACK，终于这一轮传输失败。这一轮失败之后，STA 1与STA 2采用BEB算法又一次选择随机数进行回退，可是因为两者没有办法互相监听，所以非常容易再次出现同一时候传输的现象。

所以在隐藏终端的情况下，网络性能最差时是无法传递数据包的，换言之。STA 1与STA 2的吞吐量都趋近于0。

LAN隐藏终端问题



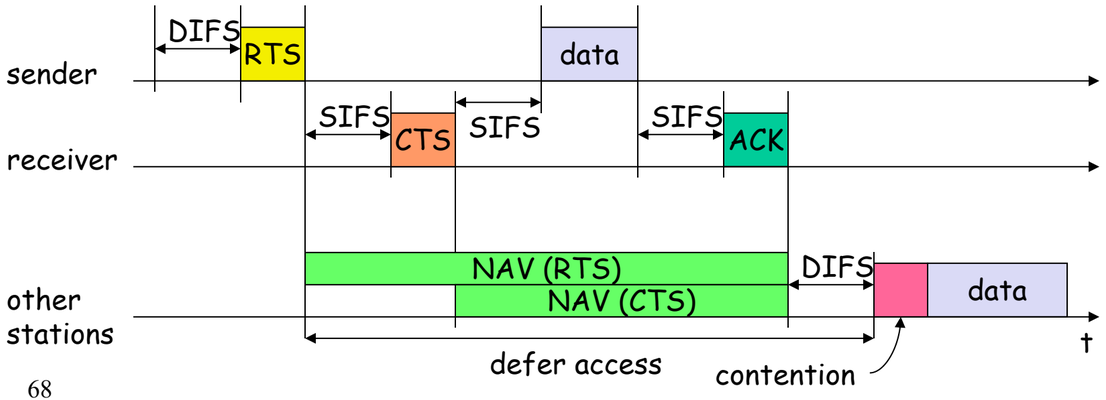
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所以在隐藏终端的情况下,网络性能最差时是无法传递数据包的,换言之。STA 1与STA 2的吞吐量都趋近于0。

RTS/CTS

802.11 CSMA/CA WITH Hidden Terminals (pearsoncmg.com)

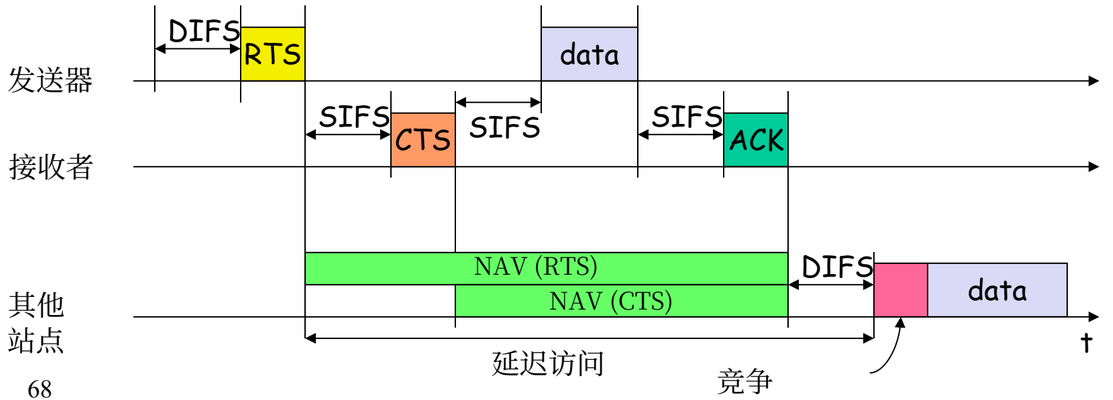
- Sending unicast packets
  - station can send RTS with reservation parameter after waiting for DIFS (reservation determines amount of time the data packet needs the medium)
  - acknowledgement via CTS after SIFS by receiver (if ready to receive)
  - other stations store medium reservations distributed via RTS and CTS
  - Two (potentially different) NAV groups



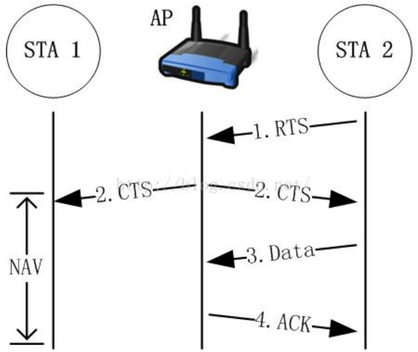
RTS/CTS

802.11 CSMA/CA 与隐藏终端（pearsoncmg.com）

- 发送单播数据包
  - 站点可以在等待 DIFS 后，使用预留参数发送 RTS（预留时间决定了数据包需要媒介的时间）
  - 接收方在SIFS后通过CTS发送确认（如果准备接收） ive)
  - 其他站点存储通过RTS和 CTS
  - 两个（可能不同的）NAV组

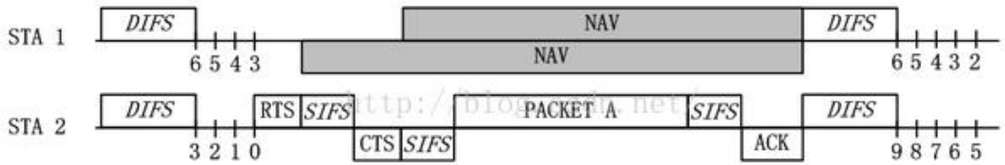


RTS/CTS



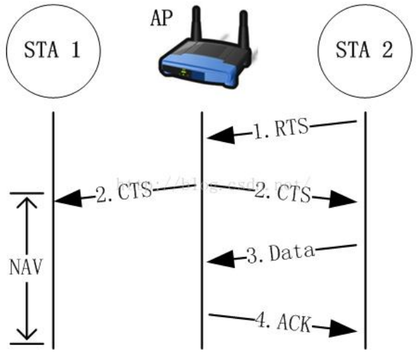
•当STA 1接收到CTS之后，该CTS不是我所请求所获得的，或者说，该CTS不是相应发给我的CTS。从而STA 1会将CTS数据帧的duration给提出。并设置在自己本地的NAV（Network Allocation Vector）上。若NAV没有倒数到0，那么其会主动悬挂其随机回退计数值，在NAV没有倒数到0之前，其随机回退计数值不再继续倒数。

当STA 2接收到CTS后。其发现该其是之前发送RTS的反馈。故节点已知信道空暇，在等待SIFS后，STA 2发送数据。当传输数据完毕之后，AP向STA 2反馈ACK，从而终于完毕一次传输。



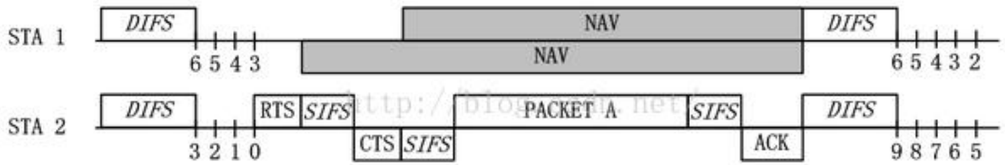
802.11协议精读2：DCF与CSMA/CA_adr5970的博客-CSDN博客

RTS/CTS



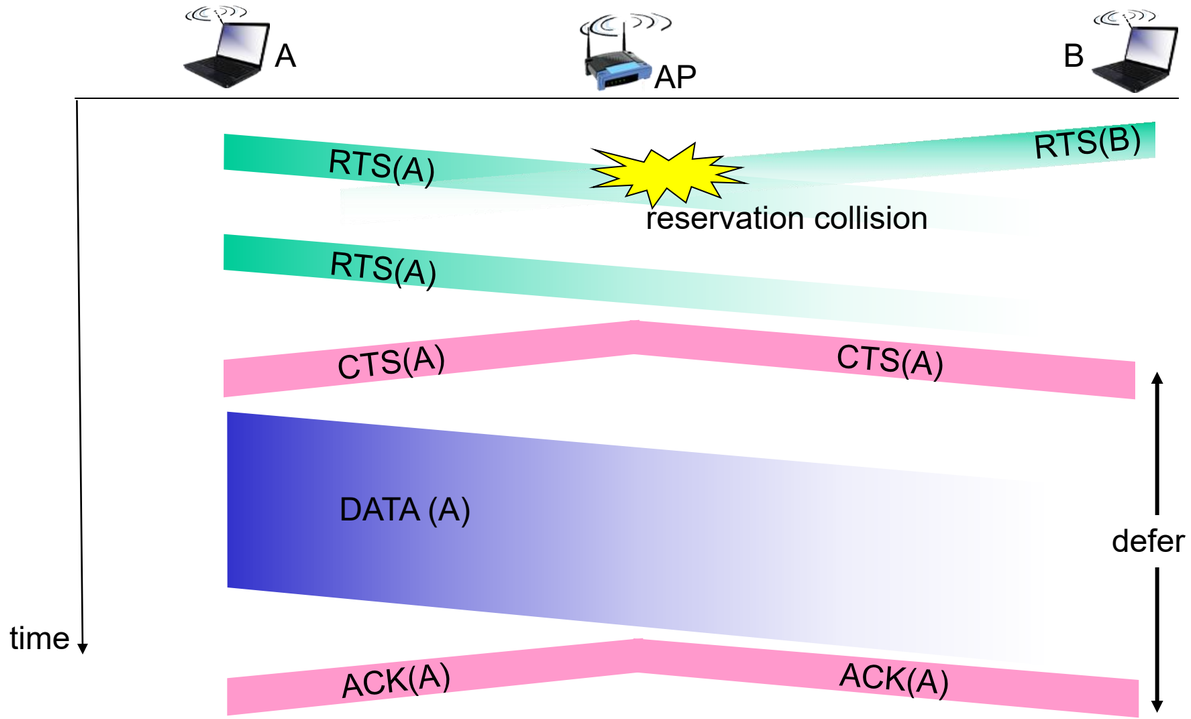
•当STA 1接收到CTS之后，该CTS不是我所请求所获得的，或者说，该CTS不是相应发给我的CTS。从而STA 1会将CTS数据帧的duration提出。并设置在自己本地的NAV（网络分配向量）上。若NAV没有倒数到0，那么其会主动悬挂其随机回退计数值，在NAV没有倒数到0之前，其随机回退计数值不再继续倒数。

当STA 2接收到CTS后。其发现该其是之前发送RTS的 反馈。故节点已知信道空暇,在等待SIFS后,STA 2发 送数据。当传输数据完毕之后,AP向STA 2反馈ACK, 从而终于完毕一次传输。

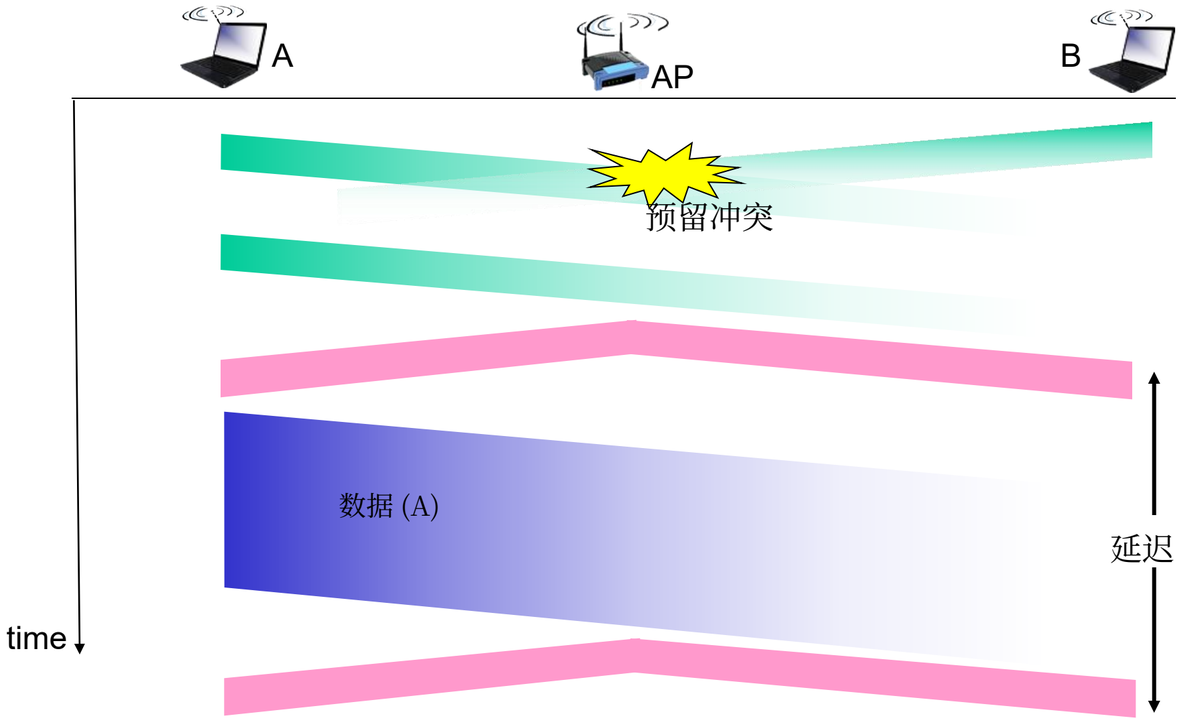


802.11协议精读2:DCF与CSMA/CA_adr5970的博客-CSDN博客

Collision Avoidance: RTS-CTS exchange



冲突避免：RTS-CTS 交换



RTS/CTS (Continued)

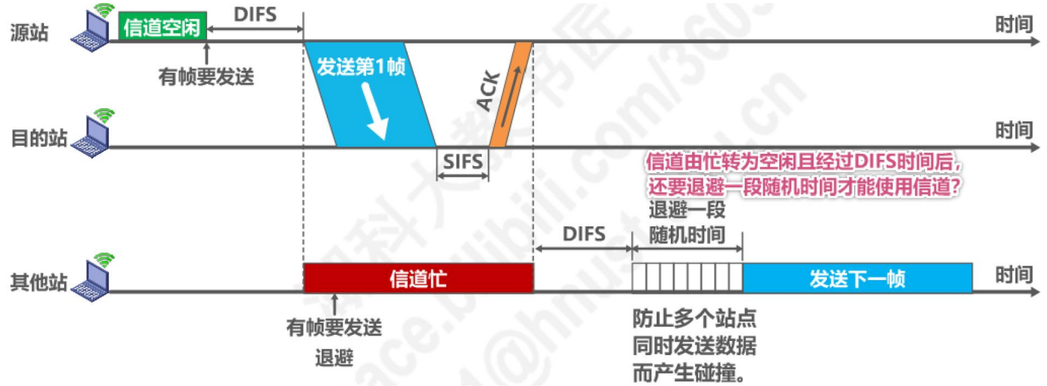
- Avoid hidden terminal problems
- Also, reduce bandwidth waste by collisions
  - Data frame can be as large as 2300bytes
  - RTS = 20bytes, CTS = 14bytes
  - The bigger is the data frame, the more advantageous
- Price : extended delay and more resource consumption!
- RTS threshold
  - Enable RTS/CTS for frames which are bigger than RTS threshold
  - Each node's decision

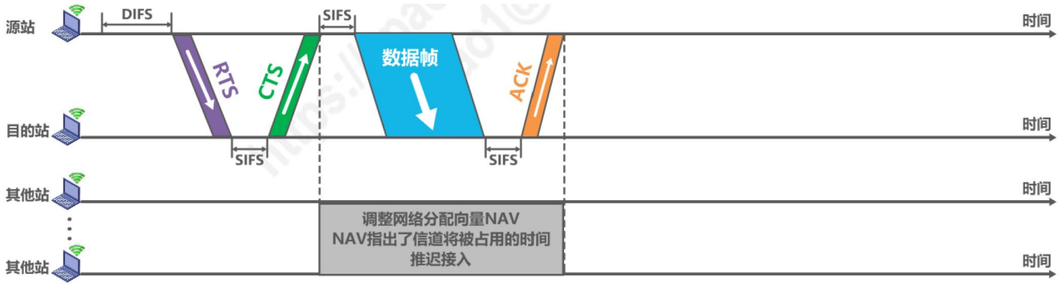
RTS/CTS (续)

- 避免隐藏终端问题
- 也通过冲突减少带宽浪费
  - 数据帧可长达 2300 字节 RTS = 20 字节, CTS = 14 字节 数据帧越大越有利
- 价格：延迟延长和资源消耗更多！
- RTS 阈值
  - 为大于 RTS 阈值的帧启用 RTS/CTS
  - 每个节点的决策

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CSMA/CA access method

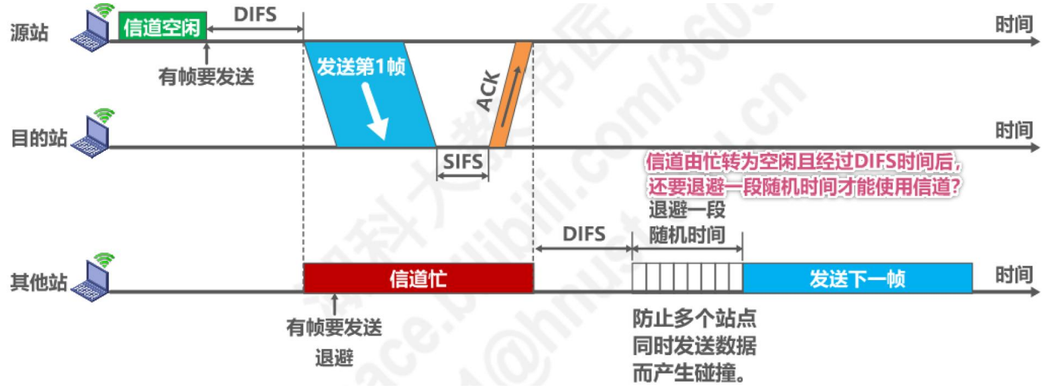


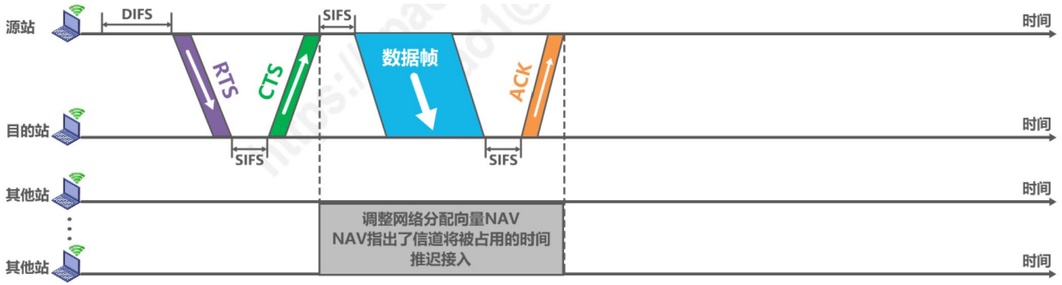


802.11 CSMA/CA WITHOUT Hidden Terminals (pearsoncmg.com)

802.11 CSMA/CA WITH Hidden Terminals (pearsoncmg.com)

CSMA/CA 访问方法



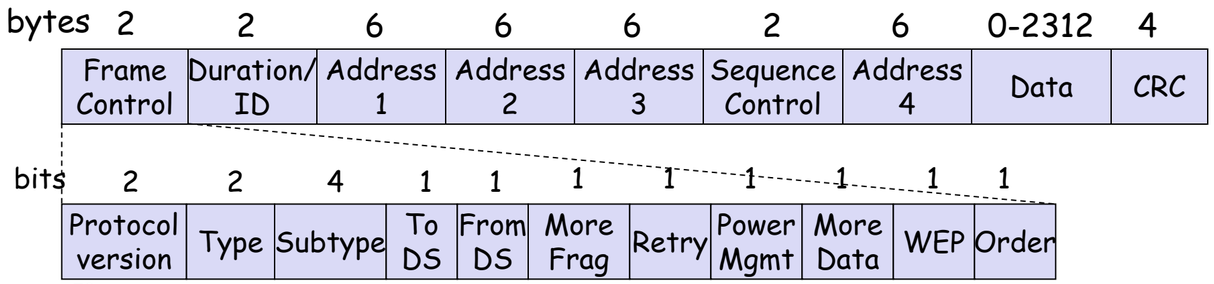


802.11 CSMA/CA 无隐藏终端 (pearsoncmg.com)

802.11 CSMA/CA 有隐藏终端 (pearsoncmg.com)

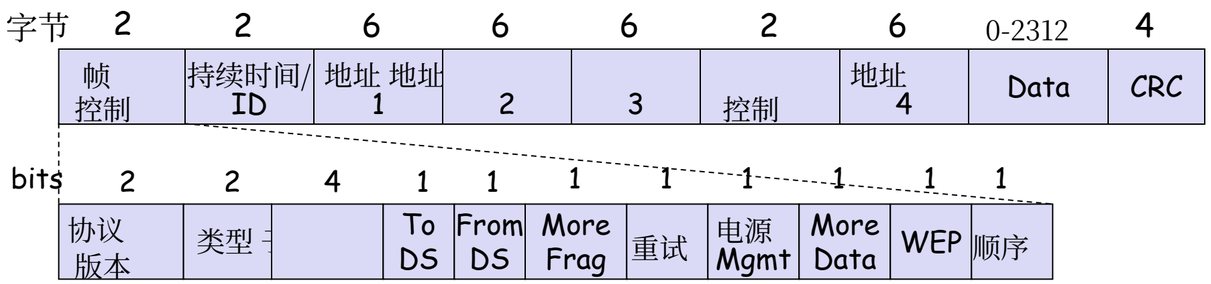
802.11 – MAC frame format

- Type/subtype
  - control frames(01), management frames(00), data frames(10)
  - E.g., data frame: type=10,subtype=0000
    beacon: type=00,subtype=1000
    RTS: type= 01,subtype=1011
- Sequence numbers
  - important against duplicated frames due to lost ACKs
- Addresses
  - receiver, transmitter (physical), BSS identifier, sender (logical)
- Miscellaneous
  - duration, checksum, frame control, data



802.11 – MAC 帧格式

- 类型/子类型
  - 控制帧(01), 管理帧(00), 数据帧(10)。 例如，数据帧：类型 =10 =0000 ,子类型信标：类型=00 =1000 ,子类型 = 01 =1011 RTS：类型 ,子类型
- 序列号
  - 重要，防止因丢失 ACK 导致的重复帧
- 地址
  - 接收器，发射器（物理），BSS 标识符，发送者（逻辑）
- 杂项。 持续时间、校验和、帧控制、数据



MAC address format

scenario	to DS	from DS	address 1	address 2	address 3	address 4
ad-hoc network	0	0	DA	SA	BSSID	-
infrastructure network, from AP	0	1	DA	BSSID	SA	-
infrastructure network, to AP	1	0	BSSID	SA	DA	-
infrastructure network, within DS	1	1	RA	TA	DA	SA

DS: Distribution System
AP: Access Point
DA: Destination Address
SA: Source Address
BSSID: Basic Service Set Identifier – MAC address of AP, cf: (E)SSID
RA: Receiver Address (AP)
TA: Transmitter Address (AP)

MAC地址格式

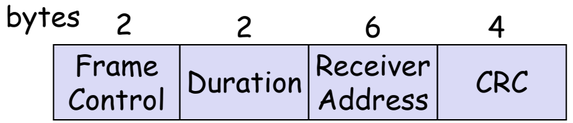
场景	从DS	DS	地址1	地址2	地址3	地址 4
自组网	0	0	DA	SA	BSSID	-
基础设施网络，来自接入点	0	1	DA	BSSID	SA	-
基础设施网络，接入点	1	0	BSSID	SA	DA	-
基础设施网络，在DS内	1	1	RA	TA	DA	SA

DS: 分布式系统 AP: 接入点 DA: 目的地址 SA: 源地址 BSSID: 基本服务集标识符 – 接入点的MAC地址， cf: (E)SSID RA: 接收器地址 (接入点) TA: 发射器地址 (接入点)

Special Frames: ACK, RTS, CTS

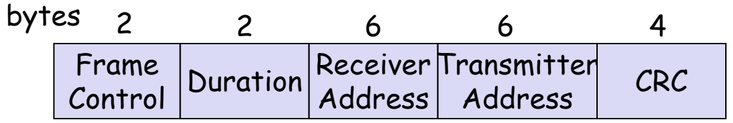
- Acknowledgement

ACK



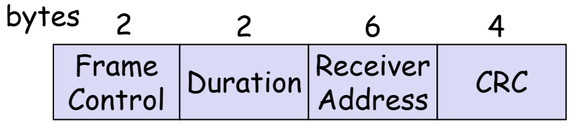
- Request To Send

RTS



- Clear To Send

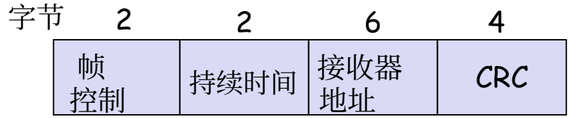
CTS



特殊帧：ACK、RTS、CTS

- 确认

ACK



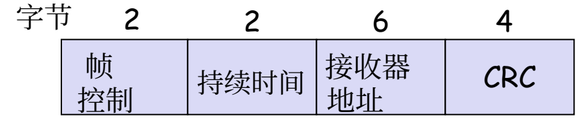
- 发送请求

RTS

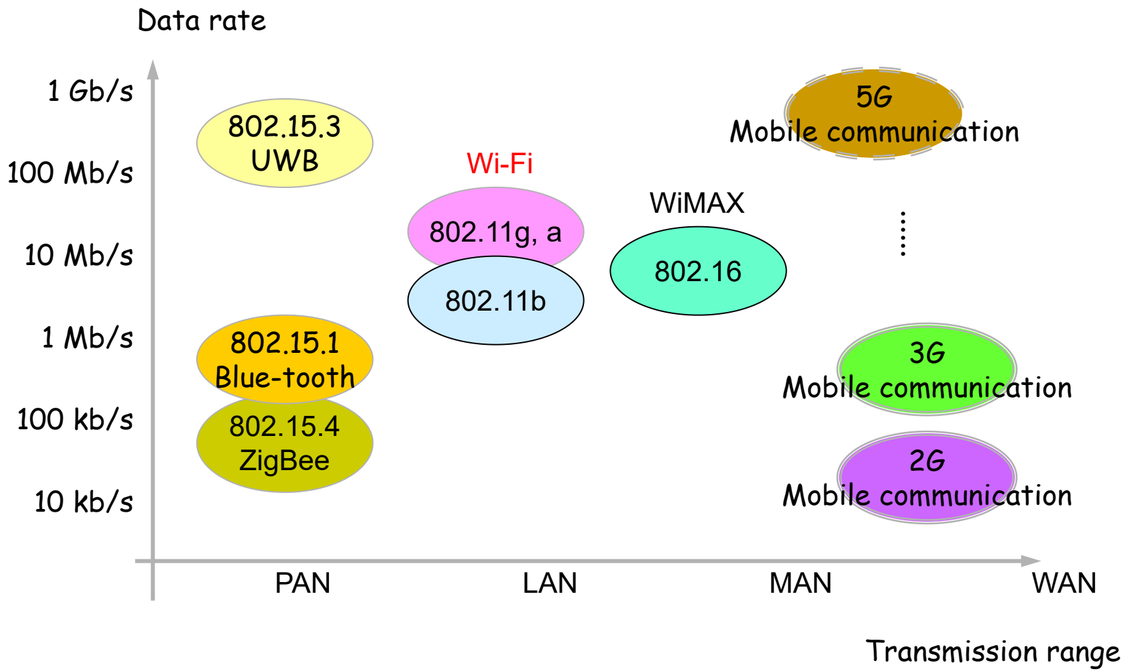


- 发送清除

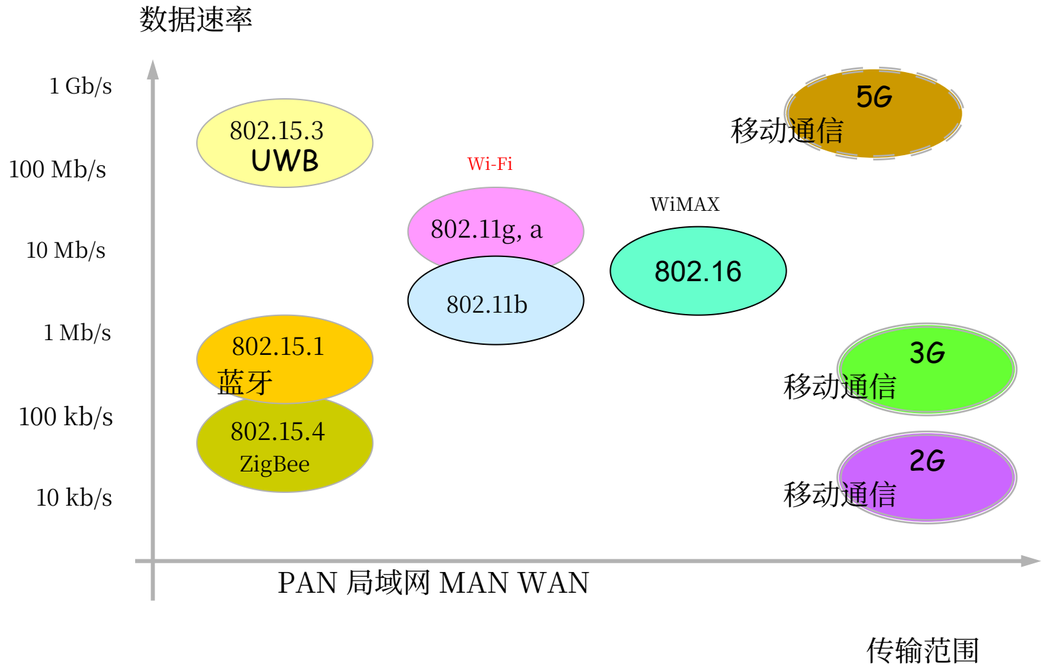
CTS



Wireless communication technology



无线通信技术



Wireless standards

Market Name	ZigBee®	---	Wi-Fi™	Bluetooth™
Standard	802.15.4	GSM/GPRS CDMA/1xRTT	802.11b	802.15.1
Application Focus	Monitoring & Control	Wide Area Voice & Data	Web, Email, Video	Cable Replacement
System Resources	4KB - 32KB	16MB+	1MB+	250KB+
Battery Life (days)	100 - 1,000+	1-7	.5 - 5	1 - 7
Network Size	Unlimited (2^{64})	1	32	7
Bandwidth (KB/s)	20 - 250	64 - 128+	11,000+	720
Transmission Range (meters)	1 - 100+	1,000+	1 - 100	1 - 10+
Success Metrics	Reliability, Power, Cost	Reach, Quality	Speed, Flexibility	Cost, Convenience

无线标准

Market Name	ZigBee®	---	Wi-Fi™	Bluetooth™
Standard	802.15.4	GSM/GPRS CDMA/1xRTT	802.11b	802.15.1
Application Focus	Monitoring & Control	Wide Area Voice & Data	Web, Email, Video	Cable Replacement
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Transmission Range (meters)	1 - 100+	1,000+	1 - 100	1 - 10+
Success Metrics	Reliability, Power, Cost	Reach, Quality	Speed, Flexibility	Cost, Convenience

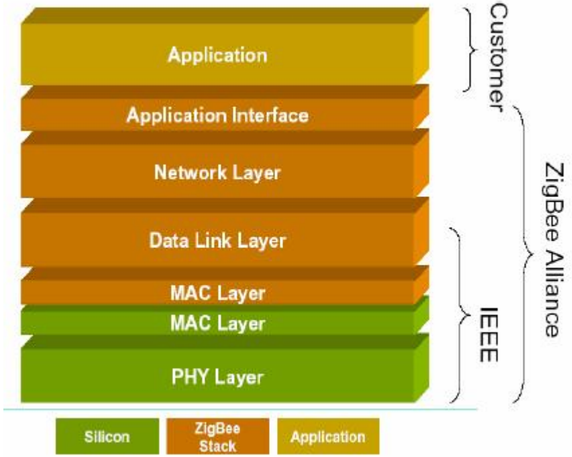
IEEE 802.15.4 and ZigBee

- IEEE 802.15.4 Working Group

  Defining lower layers of protocol stack: MAC and PHY

- ZigBee Alliance

  Defining upper layers of protocol stack from network to application



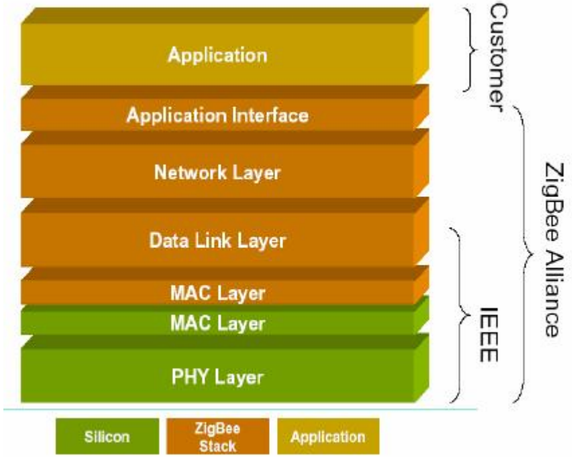
IEEE 802.15.4 和 ZigBee

- IEEE 802.15.4 工作组

  定义协议栈的较低层：MAC 和 物理层

- ZigBee 联盟

  从网络到应用定义协议栈的上层



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ZigBee overview

- ZigBee was created to address the market need for a cost-effective, standards based wireless networking solution that supports low data-rates, low-power consumption, security, and reliability.
- ZigBee is the only standards-based technology that addresses the unique needs of most remote monitoring and control and sensory network applications.
- The initial markets for the ZigBee Alliance include Home Automation, Building Automation and Industrial Automation.

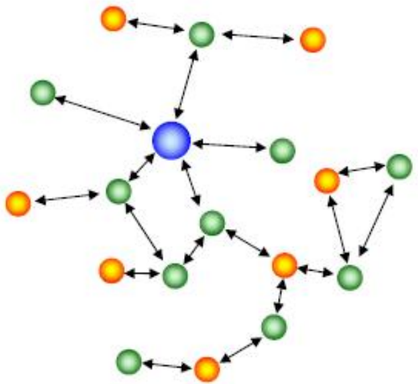
ZigBee 概述

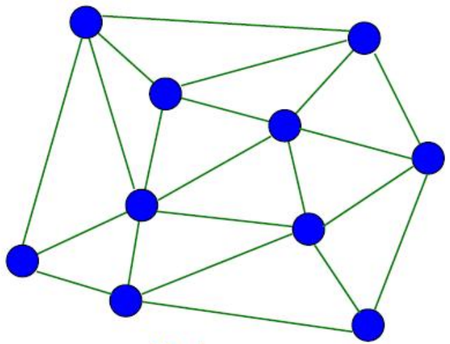
- ZigBee是为了应对市场对低成本、标准化无线网络解决方案的需求而创建的，该方案支持低数据速率、低功耗、安全性和可靠性。
- ZigBee是唯一基于标准的技术，能够满足大多数 远程监控和控制 以及 传感器网络应用的特殊需求。
- ZigBee 联盟的初始市场包括家庭自动化、楼宇自动化和工业自动化。

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How to achieve low power consumption?

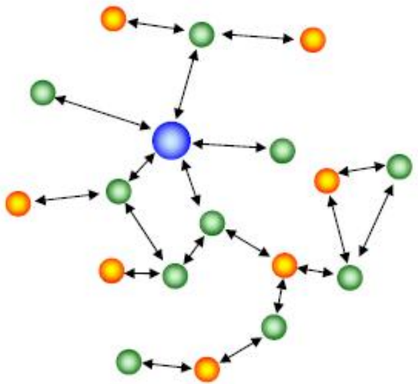
- The duty cycle of battery is designed to be very low, resulting in very low average power consumption.
- Once associated with a network, a ZigBee node can wake up and communicate with other devices and return to sleep.
- Short range operation.
- Simple but flexible protocol.

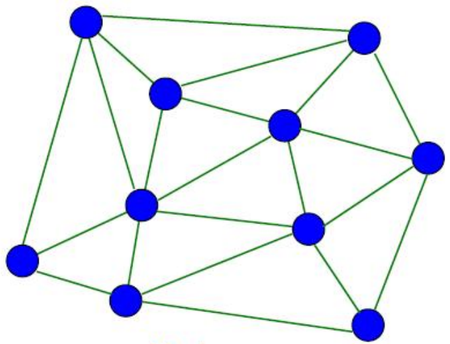




如何实现低功耗 ?

- 电源 的充放电周期 被设计为 非常低 ， 导致平均电源消耗非常低。
- 一旦与网络关联，ZigBee 节点可以 唤醒 并与其他设备通信，然后返回到睡眠状态。
- 短距离 操作。
- 简单但灵活的协议。



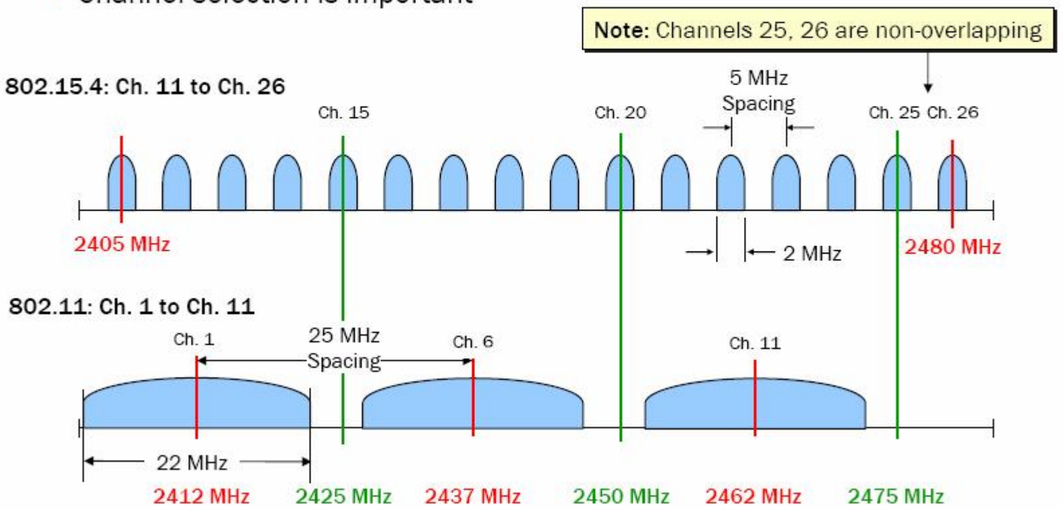


Interference and Coexistence in the 2.4GHz Band

802.15.4 and 802.11b Spectrum Relationship

Co-exists with WiFi, Bluetooth

- Channel selection is important

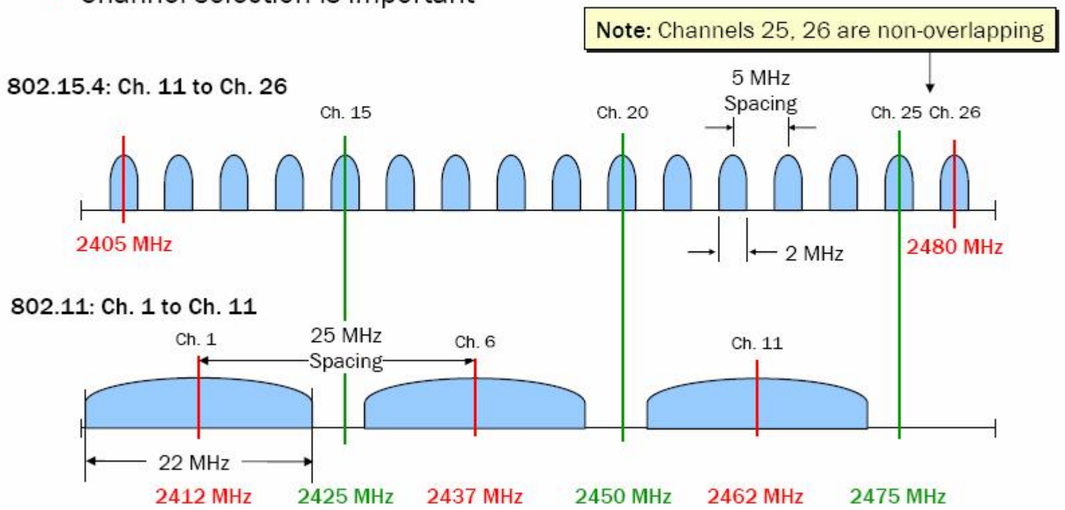


干扰与共存于2.4GHz频段

802.15.4 and 802.11b Spectrum Relationship

Co-exists with WiFi, Bluetooth

- Channel selection is important



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Chapter 4:Local area network Summary

- various link layer technologies
  - LAN model
  - Ethernet
  - hubs, switches
  - VLAN
  - IEEE 802.11
  - IEEE 802.15

802.11协议精读

第四章：局域网摘要

- 各种链路层技术
  - 局域网模型
  - 以太网
  - 集线器、交换机
  - 虚拟局域网
  - IEEE 802.11
  - IEEE 802.15

802.11协议精读